

Titration report



Method

Method ID	SERCALKSAMPLES	Release state	Released
Name	TA of Seawater	Modified by	SystemInternal
Modified on	2/22/2024 12:07:06 PM		

Summary

Samples

	Sample ID	Name	Result	State
1	FTS2023 8/16 14:30rep3, 1	Sample size	63.219 g	
		Content R1	2.1992 mmol/L	OK
2	FTS2023 8/16 14:30rep4, 2	Sample size	59.875 g	
		Content R1	2.1390 mmol/L ⁽⁹⁾	OK
3	FTS2023 8/8 5:30rep3, 3	Sample size	55.714 g	
		Content R1	1.3546 mmol/L	OK
4	FTS2023 8/8 5:30rep4, 4	Sample size	52.704 g	
		Content R1	1.3479 mmol/L ⁽¹⁾	OK
5	FTS2023 8/8 7:30rep3, 5	Sample size	63.672 g	
		Content R1	1.0368 mmol/L	OK
6	FTS2023 8/8 0830 rep3, 6	Sample size	61.527 g	
		Content R1	1.0827 mmol/L	OK
7	FTS2023 8/8 0930rep3, 7	Sample size	64.432 g	
		Content R1	1.3784 mmol/L ⁽⁹⁾	OK
8	FTS2023 8/81030 rep3, 8	Sample size	60.589 g	
		Content R1	1.4660 mmol/L ⁽⁹⁾	OK
9	FTS2023 8/81030 rep4, 9	Sample size	64.881 g	
		Content R1	1.4709 mmol/L ⁽¹⁾	OK
10	FTS2023 8/8 1230 rep3, 10	Sample size	56.151 g	
		Content R1	1.6967 mmol/L ⁽⁹⁾	OK
11	FTS2023 8/8 1230 rep4, 11	Sample size	41.047 g	
		Content R1	1.7260 mmol/L	OK
12	FTS2023 8/8 1430 rep3, 12	Sample size	56.546 g	
		Content R1	1.9364 mmol/L ⁽⁹⁾	OK
13	FTS2023 8/8 1430 rep4, 13	Sample size	60.100 g	
		Content R1	1.9334 mmol/L ⁽⁹⁾	OK
14	FTS2023 8/8 1530 rep3, 14	Sample size	57.438 g	
		Content R1	1.9449 mmol/L ⁽⁹⁾	OK
15	FTS2023 8/8 1530 rep4, 15	Sample size	63.249 g ⁽¹²⁾	

Statistics

R1

Minimum	1.6224 mmol/L ^(1·9)	Minimum	1.0368 mmol/L
Standard deviation	0.3725 ^(1·9)	Rel. standard deviation	22.958 % ^(1·9)
Total samples	14	Samples excluded	0

Titration report



Main Result Details

Result title	Content	Sample ID1	FTS2023 8/16 14:30rep3
Result name	R1	Sample ID2	1
Result	2.1992 mmol/L	Result comment	
		Result state	OK
		Result state comment	
		Nominal value	
		Compliance state	Released
		Is result deleted	No

Calculation formula R1=Q*C/m

Result title	Content	Sample ID1	FTS2023 8/16 14:30rep4
Result name	R1	Sample ID2	2
Result	2.1390 mmol/L ⁽⁹⁾	Result comment	
		Result state	OK
		Result state comment	
		Nominal value	
		Compliance state	Released
		Is result deleted	No

Calculation formula R1=Q*C/m

Result title	Content	Sample ID1	FTS2023 8/8 5:30rep3
Result name	R1	Sample ID2	3
Result	1.3546 mmol/L	Result comment	
		Result state	OK
		Result state comment	
		Nominal value	
		Compliance state	Released
		Is result deleted	No

Calculation formula R1=Q*C/m

Result title	Content	Sample ID1	FTS2023 8/8 5:30rep4
Result name	R1	Sample ID2	4
Result	1.3479 mmol/L ⁽¹⁾	Result comment	
		Result state	OK
		Result state comment	
		Nominal value	
		Compliance state	Released
		Is result deleted	No

Calculation formula R1=Q*C/m

Result title	Content	Sample ID1	FTS2023 8/8 7:30rep3
Result name	R1	Sample ID2	5
Result	1.0368 mmol/L	Result comment	

Titration report



Result state	OK
Result state comment	
Nominal value	
Compliance state	Released
Is result deleted	No

Calculation formula R1=Q*C/m

Result title	Content
Result name	R1
Result	1.0827 mmol/L

Sample ID1	FTS2023 8/8 0830 rep3
Sample ID2	6
Result comment	
Result state	OK
Result state comment	
Nominal value	
Compliance state	Released
Is result deleted	No

Calculation formula R1=Q*C/m

Result title	Content
Result name	R1
Result	1.3784 mmol/L ⁽⁹⁾

Sample ID1	FTS2023 8/8 0930rep3
Sample ID2	7
Result comment	
Result state	OK
Result state comment	
Nominal value	
Compliance state	Released
Is result deleted	No

Calculation formula R1=Q*C/m

Result title	Content
Result name	R1
Result	1.4660 mmol/L ⁽⁹⁾

Sample ID1	FTS2023 8/81030 rep3
Sample ID2	8
Result comment	
Result state	OK
Result state comment	
Nominal value	
Compliance state	Released
Is result deleted	No

Calculation formula R1=Q*C/m

Result title	Content
Result name	R1
Result	1.4709 mmol/L ⁽¹⁾

Sample ID1	FTS2023 8/81030 rep4
Sample ID2	9
Result comment	
Result state	OK
Result state comment	
Nominal value	
Compliance state	Released

Titration report

LabX

Is result deleted No

Calculation formula $R1=Q \cdot C/m$

Result title	Content	Sample ID1	FTS2023 8/8 1230 rep3
Result name	R1	Sample ID2	10
Result	1.6967 mmol/L ⁽⁹⁾	Result comment	
		Result state	OK
		Result state comment	
		Nominal value	
		Compliance state	Released
		Is result deleted	No

Calculation formula $R1=Q \cdot C/m$

Result title	Content	Sample ID1	FTS2023 8/8 1230 rep4
Result name	R1	Sample ID2	11
Result	1.7260 mmol/L	Result comment	
		Result state	OK
		Result state comment	
		Nominal value	
		Compliance state	Released
		Is result deleted	No

Calculation formula $R1=Q \cdot C/m$

Result title	Content	Sample ID1	FTS2023 8/8 1430 rep3
Result name	R1	Sample ID2	12
Result	1.9364 mmol/L ⁽⁹⁾	Result comment	
		Result state	OK
		Result state comment	
		Nominal value	
		Compliance state	Released
		Is result deleted	No

Calculation formula $R1=Q \cdot C/m$

Result title	Content	Sample ID1	FTS2023 8/8 1430 rep4
Result name	R1	Sample ID2	13
Result	1.9334 mmol/L ⁽⁹⁾	Result comment	
		Result state	OK
		Result state comment	
		Nominal value	
		Compliance state	Released
		Is result deleted	No

Calculation formula $R1=Q \cdot C/m$

Result title	Content	Sample ID1	FTS2023 8/8 1530 rep3
--------------	---------	------------	-----------------------

Titration report

LabX

Result name	R1	Sample ID2	14
Result	1.9449 mmol/L ⁽⁹⁾	Result comment	
		Result state	OK
		Result state comment	
		Nominal value	
		Compliance state	Released
		Is result deleted	No

Calculation formula R1=Q*C/m

Raw Results

Scope 1/1, Sample 1/17 (FTS2023 8/16 14:30rep3; 1)

Sample (S1)

Creation time	2/22/2024 12:10:06 PM
Sample start time	2/22/2024 12:10:06 PM
Sample size	63.219 g
Correction factor	1.0
Temperature	25.0 °C
Density	1.0 g/mL

Titration stand (Titrationstand1)

Creation time	2/22/2024 12:10:06 PM
---------------	-----------------------

Measure (normal) [1] (Measurenormal1)

Creation time	2/22/2024 12:10:07 PM
T	25.0 °C
E	7.792 pH
tMe	01:15 min
ESTMe	4.014 pH

Titration (EQP) [1] (TitrationEQP1)

Creation time	2/22/2024 12:11:22 PM
Terminated at	Potential
EEQ1	4.256 pH
EHNV1	6.209 pH
Q1	0.139031 mmol
VEQ1	1.388923 mL
EST	7.801 pH
VEX	0.791327 mL
VEND	2.18025 mL
t	24:53 min
QEND	0.218243 mmol
QEX	0.079212 mmol
EPD	6.743 pH
nEQ	1

Content (R1)

Titration report

LabX

Creation time	2/22/2024 12:36:20 PM
Result	2.1992 mmol/L
Decimal places	4
Substance	None
M	1 g/mol
z	1

Rinse (Rinse1)

Creation time	2/22/2024 12:36:20 PM
---------------	-----------------------

Scope 1/1, Sample 2/17 (FTS2023 8/16 14:30rep4; 2)

Sample (S1)

Creation time	2/22/2024 12:36:31 PM
Sample start time	2/22/2024 12:36:31 PM
Sample size	59.875 g
Correction factor	1.0
Temperature	25.0 °C
Density	1.0 g/mL

Titration stand (Titrationstand1)

Creation time	2/22/2024 12:36:32 PM
---------------	-----------------------

Measure (normal) [1] (Measurenormal1)

Creation time	2/22/2024 12:36:32 PM
T	25.2 °C
E	7.872 pH
tMe	01:12 min
ESTMe	4.644 pH

Titration (EQP) [1] (TitrationEQP1)

Creation time	2/22/2024 12:37:45 PM
Terminated at	Potential
EEQ1	4.557 pH ⁽⁹⁾
EHN1	6.241 pH
Q1	0.128073 mmol ⁽⁹⁾
VEQ1	1.279453 mL ⁽⁹⁾
EST	7.877 pH
VEX	0.778797 mL
VEND	2.05825 mL
t	23:53 min
QEND	0.206031 mmol
QEX	0.077958 mmol
EPD	6.727 pH
nEQ	1

Content (R1)

Creation time	2/22/2024 1:01:43 PM ⁽⁹⁾
Result	2.1390 mmol/L ⁽⁹⁾
Decimal places	4 ⁽⁹⁾
Substance	None ⁽⁹⁾

Titration report

LabX

M 1 g/mol ⁽⁹⁾
z 1 ⁽⁹⁾

Rinse (Rinse1)

Creation time 2/22/2024 1:01:43 PM

Scope 1/1, Sample 3/17 (FTS2023 8/8 5:30rep3; 3)

Sample (S1)

Creation time 2/22/2024 1:01:54 PM
Sample start time 2/22/2024 1:01:54 PM
Sample size 55.714 g
Correction factor 1.0
Temperature 25.0 °C
Density 1.0 g/mL

Titration stand (Titrationstand1)

Creation time 2/22/2024 1:01:54 PM

Measure (normal) [1] (Measurenormal1)

Creation time 2/22/2024 1:01:54 PM
T 25.2 °C
E 7.414 pH
tMe 01:12 min
ESTMe 4.593 pH

Titration (EQP) [1] (TitrationEQP1)

Creation time 2/22/2024 1:03:07 PM
Terminated at Potential
EEQ1 4.601 pH
EHNV1 6.226 pH
Q1 0.075468 mmol
VEQ1 0.753922 mL
EST 7.435 pH
VEX 0.749078 mL
VEND 1.50300 mL
t 19:47 min
QEND 0.150450 mmol
QEX 0.074983 mmol
EPD 6.372 pH
nEQ 2

Content (R1)

Creation time 2/22/2024 1:23:00 PM
Result 1.3546 mmol/L
Decimal places 4
Substance None
M 1 g/mol
z 1

Rinse (Rinse1)

Titration report

LabX

Creation time 2/22/2024 1:23:00 PM

Scope 1/1, Sample 4/17 (FTS2023 8/8 5:30rep4; 4)

Sample (S1)

Creation time 2/22/2024 1:23:11 PM
Sample start time 2/22/2024 1:23:11 PM
Sample size 52.704 g
Correction factor 1.0
Temperature 25.0 °C
Density 1.0 g/mL

Titration stand (Titrationstand1)

Creation time 2/22/2024 1:23:12 PM

Measure (normal) [1] (Measurenormal1)

Creation time 2/22/2024 1:23:12 PM
T 25.2 °C
E 7.488 pH
tMe 01:12 min
ESTMe 4.511 pH

Titration (EQP) [1] (TitrationEQP1)

Creation time 2/22/2024 1:24:25 PM
Terminated at Potential
EEQ1 4.652 pH
EHN1 6.230 pH
Q1 0.071040 mmol
VEQ1 0.709694 mL
EST 7.507 pH
VEX 0.698679 mL
VEND 1.42825 mL
t 19:19 min
QEND 0.142968 mmol
QEX 0.069938 mmol
EPD 6.338 pH
nEQ 2 ⁽²²⁾

Content (R1)

Creation time 2/22/2024 1:43:49 PM ⁽¹⁾
Result 1.3479 mmol/L ⁽¹⁾
Decimal places 4 ⁽¹⁾
Substance None ⁽¹⁾
M 1 g/mol ⁽¹⁾
z 1 ⁽¹⁾

Rinse (Rinse1)

Creation time 2/22/2024 1:43:49 PM

Scope 1/1, Sample 5/17 (FTS2023 8/8 7:30rep3; 5)

Sample (S1)

Titration report

LabX

Creation time	2/22/2024 1:44:00 PM
Sample start time	2/22/2024 1:44:00 PM
Sample size	63.672 g
Correction factor	1.0
Temperature	25.0 °C
Density	1.0 g/mL

Titration stand (Titrationstand1)

Creation time	2/22/2024 1:44:00 PM
----------------------	----------------------

Measure (normal) [1] (Measurenormal1)

Creation time	2/22/2024 1:44:00 PM
T	25.2 °C
E	7.220 pH
tMe	01:12 min
ESTMe	4.629 pH

Titration (EQP) [1] (TitrationEQP1)

Creation time	2/22/2024 1:45:13 PM
Terminated at	Potential
EEQ1	4.702 pH
EHNV1	6.214 pH
Q1	0.066013 mmol
VEQ1	0.659466 mL
EST	7.244 pH
VEX	0.813784 mL
VEND	1.47325 mL
t	19:47 min
QEND	0.147472 mmol
QEX	0.081460 mmol
EPD	6.275 pH
nEQ	2

Content (R1)

Creation time	2/22/2024 2:05:06 PM
Result	1.0368 mmol/L
Decimal places	4
Substance	None
M	1 g/mol
z	1

Rinse (Rinse1)

Creation time	2/22/2024 2:05:06 PM
----------------------	----------------------

Scope 1/1, Sample 6/17 (FTS2023 8/8 0830 rep3; 6)

Sample (S1)

Creation time	2/22/2024 2:05:17 PM
Sample start time	2/22/2024 2:05:17 PM
Sample size	61.527 g
Correction factor	1.0

Titration report

LabX

Temperature 25.0 °C
Density 1.0 g/mL

Titration stand (Titrationstand1)

Creation time 2/22/2024 2:05:17 PM

Measure (normal) [1] (Measurenormal1)

Creation time 2/22/2024 2:05:18 PM
T 25.3 °C
E 7.266 pH
tMe 01:12 min
ESTMe 4.604 pH

Titration (EQP) [1] (TitrationEQP1)

Creation time 2/22/2024 2:06:30 PM
Terminated at Potential
EEQ1 4.647 pH
EHN1 6.210 pH
Q1 0.066617 mmol
VEQ1 0.665500 mL
EST 7.290 pH
VEX 0.800500 mL
VEND 1.46600 mL
t 19:46 min
QEND 0.146747 mmol
QEX 0.080130 mmol
EPD 6.277 pH
nEQ 2

Content (R1)

Creation time 2/22/2024 2:26:22 PM
Result 1.0827 mmol/L
Decimal places 4
Substance None
M 1 g/mol
z 1

Rinse (Rinse1)

Creation time 2/22/2024 2:26:23 PM

Scope 1/1, Sample 7/17 (FTS2023 8/8 0930rep3; 7)

Sample (S1)

Creation time 2/22/2024 2:26:34 PM
Sample start time 2/22/2024 2:26:34 PM
Sample size 64.432 g
Correction factor 1.0
Temperature 25.0 °C
Density 1.0 g/mL

Titration stand (Titrationstand1)

Titration report

LabX

Creation time 2/22/2024 2:26:34 PM

Measure (normal) [1] (Measurenormal1)

Creation time 2/22/2024 2:26:34 PM
T 25.3 °C
E 7.426 pH
tMe 01:12 min
ESTMe 4.686 pH

Titration (EQP) [1] (TitrationEQP1)

Creation time 2/22/2024 2:27:48 PM
Terminated at Potential
EEQ1 4.616 pH ⁽⁹⁾
EHN1 6.221 pH
Q1 0.088814 mmol ⁽⁹⁾
VEQ1 0.887257 mL ⁽⁹⁾
EST 7.447 pH
VEX 0.835493 mL
VEND 1.72275 mL
t 21:18 min
QEND 0.172447 mmol
QEX 0.083633 mmol
EPD 6.472 pH
nEQ 2 ⁽⁹⁾

Content (R1)

Creation time 2/22/2024 2:49:11 PM ⁽⁹⁾
Result 1.3784 mmol/L ⁽⁹⁾
Decimal places 4 ⁽⁹⁾
Substance None ⁽⁹⁾
M 1 g/mol ⁽⁹⁾
z 1 ⁽⁹⁾

Rinse (Rinse1)

Creation time 2/22/2024 2:49:11 PM

Scope 1/1, Sample 8/17 (FTS2023 8/81030 rep3; 8)

Sample (S1)

Creation time 2/22/2024 2:49:23 PM
Sample start time 2/22/2024 2:49:23 PM
Sample size 60.589 g
Correction factor 1.0
Temperature 25.0 °C
Density 1.0 g/mL

Titration stand (Titrationstand1)

Creation time 2/22/2024 2:49:23 PM

Measure (normal) [1] (Measurenormal1)

Creation time 2/22/2024 2:49:23 PM

Titration report

LabX

T	25.3 °C
E	7.374 pH
tMe	01:12 min
ESTMe	4.715 pH

Titration (EQP) [1] (TitrationEQP1)

Creation time	2/22/2024 2:50:36 PM
Terminated at	Potential
EEQ1	4.638 pH ⁽⁹⁾
EHN1	6.211 pH
Q1	0.088822 mmol ⁽⁹⁾
VEQ1	0.887330 mL ⁽⁹⁾
EST	7.391 pH
VEX	0.788920 mL
VEND	1.67625 mL
t	20:48 min
QEND	0.167793 mmol
QEX	0.078971 mmol
EPD	6.467 pH
nEQ	2 ⁽⁹⁾

Content (R1)

Creation time	2/22/2024 3:11:29 PM ⁽⁹⁾
Result	1.4660 mmol/L ⁽⁹⁾
Decimal places	4 ⁽⁹⁾
Substance	None ⁽⁹⁾
M	1 g/mol ⁽⁹⁾
z	1 ⁽⁹⁾

Rinse (Rinse1)

Creation time	2/22/2024 3:11:29 PM
---------------	----------------------

Scope 1/1, Sample 9/17 (FTS2023 8/81030 rep4; 9)

Sample (S1)

Creation time	2/22/2024 3:11:40 PM
Sample start time	2/22/2024 3:11:40 PM
Sample size	64.881 g
Correction factor	1.0
Temperature	25.0 °C
Density	1.0 g/mL

Titration stand (Titrationstand1)

Creation time	2/22/2024 3:11:41 PM
---------------	----------------------

Measure (normal) [1] (Measurenormal1)

Creation time	2/22/2024 3:11:41 PM
T	25.3 °C
E	7.416 pH
tMe	01:12 min
ESTMe	4.755 pH

Titration report

LabX

Titration (EQP) [1] (TitrationEQP1)

Creation time	2/22/2024 3:12:54 PM
Terminated at	Potential
EEQ1	4.601 pH
EHN1	6.218 pH
Q1	0.095435 mmol
VEQ1	0.953395 mL
EEQ2	4.553 pH
EHN2	6.213 pH
Q2	0.095937 mmol
VEQ2	0.958408 mL
EST	7.435 pH
VEX	0.837842 mL
VEND	1.79625 mL
t	21:48 min
QEND	0.179805 mmol
QEX	0.083868 mmol
EPD	6.514 pH
nEQ	2

Content (R1)

Creation time	2/22/2024 3:34:47 PM ⁽¹⁾
Result	1.4709 mmol/L ⁽¹⁾
Decimal places	4 ⁽¹⁾
Substance	None ⁽¹⁾
M	1 g/mol ⁽¹⁾
z	1 ⁽¹⁾

Rinse (Rinse1)

Creation time	2/22/2024 3:34:48 PM
----------------------	----------------------

Scope 1/1, Sample 10/17 (FTS2023 8/8 1230 rep3; 10)

Sample (S1)

Creation time	2/22/2024 3:34:59 PM
Sample start time	2/22/2024 3:34:59 PM
Sample size	56.151 g
Correction factor	1.0
Temperature	25.0 °C
Density	1.0 g/mL

Titration stand (Titrationstand1)

Creation time	2/22/2024 3:34:59 PM
----------------------	----------------------

Measure (normal) [1] (Measurenormal1)

Creation time	2/22/2024 3:34:59 PM
T	25.3 °C
E	7.657 pH
tMe	01:12 min
ESTMe	4.723 pH

Titration report

LabX

Titration (EQP) [1] (TitrationEQP1)

Creation time	2/22/2024 3:36:12 PM
Terminated at	Potential
EEQ1	4.586 pH ⁽⁹⁾
EHN1	6.237 pH
Q1	0.095270 mmol ⁽⁹⁾
VEQ1	0.951747 mL ⁽⁹⁾
EST	7.669 pH
VEX	0.764003 mL
VEND	1.71575 mL
t	21:22 min
QEND	0.171747 mmol
QEX	0.076477 mmol
EPD	6.535 pH
nEQ	2 ⁽⁹⁾

Content (R1)

Creation time	2/22/2024 3:57:40 PM ⁽⁹⁾
Result	1.6967 mmol/L ⁽⁹⁾
Decimal places	4 ⁽⁹⁾
Substance	None ⁽⁹⁾
M	1 g/mol ⁽⁹⁾
z	1 ⁽⁹⁾

Rinse (Rinse1)

Creation time	2/22/2024 3:57:40 PM
----------------------	----------------------

Scope 1/1, Sample 11/17 (FTS2023 8/8 1230 rep4; 11)

Sample (S1)

Creation time	2/22/2024 3:57:51 PM
Sample start time	2/22/2024 3:57:51 PM
Sample size	41.047 g
Correction factor	1.0
Temperature	25.0 °C
Density	1.0 g/mL

Titration stand (Titrationstand1)

Creation time	2/22/2024 3:57:51 PM
----------------------	----------------------

Measure (normal) [1] (Measurenormal1)

Creation time	2/22/2024 3:57:51 PM
T	25.3 °C
E	7.781 pH
tMe	01:12 min
ESTMe	4.485 pH

Titration (EQP) [1] (TitrationEQP1)

Creation time	2/22/2024 3:59:05 PM
Terminated at	Potential

Titration report

LabX

EEQ2	4.609 pH
EHN2	6.282 pH
Q2	0.071184 mmol
VEQ2	0.711124 mL
EEQ1	4.661 pH
EHN1	6.286 pH
Q1	0.070846 mmol
VEQ1	0.707748 mL
EST	7.790 pH
VEX	0.553752 mL
VEND	1.26150 mL
t	17:54 min
QEND	0.126276 mmol
QEX	0.055431 mmol
EPD	6.385 pH
nEQ	2

Content (R1)

Creation time	2/22/2024 4:17:04 PM
Result	1.7260 mmol/L
Decimal places	4
Substance	None
M	1 g/mol
z	1

Rinse (Rinse1)

Creation time	2/22/2024 4:17:05 PM
---------------	----------------------

Scope 1/1, Sample 12/17 (FTS2023 8/8 1430 rep3; 12)

Sample (S1)

Creation time	2/22/2024 4:17:16 PM
Sample start time	2/22/2024 4:17:16 PM
Sample size	56.546 g
Correction factor	1.0
Temperature	25.0 °C
Density	1.0 g/mL

Titration stand (Titrationstand1)

Creation time	2/22/2024 4:17:16 PM
---------------	----------------------

Measure (normal) [1] (Measurenormal1)

Creation time	2/22/2024 4:17:16 PM
T	25.1 °C
E	7.683 pH
tMe	13:53 min
ESTMe	3.573 pH

Titration (EQP) [1] (TitrationEQP1)

Creation time	2/22/2024 4:31:10 PM
Terminated at	Potential

Titration report

LabX

EEQ1	4.560 pH ⁽⁹⁾
EHN1	6.224 pH
Q1	0.109493 mmol ⁽⁹⁾
VEQ1	1.093841 mL ⁽⁹⁾
EST	7.692 pH
VEX	0.775159 mL
VEND	1.86900 mL
t	22:21 min
QEND	0.187087 mmol
QEX	0.077593 mmol
EPD	6.608 pH
nEQ	2 ⁽⁹⁾

Content (R1)

Creation time	2/22/2024 4:53:36 PM ⁽⁹⁾
Result	1.9364 mmol/L ⁽⁹⁾
Decimal places	4 ⁽⁹⁾
Substance	None ⁽⁹⁾
M	1 g/mol ⁽⁹⁾
z	1 ⁽⁹⁾

Rinse (Rinse1)

Creation time	2/22/2024 4:53:36 PM
---------------	----------------------

Scope 1/1, Sample 13/17 (FTS2023 8/8 1430 rep4; 13)

Sample (S1)

Creation time	2/22/2024 4:53:48 PM
Sample start time	2/22/2024 4:53:48 PM
Sample size	60.100 g
Correction factor	1.0
Temperature	25.0 °C
Density	1.0 g/mL

Titration stand (Titrationstand1)

Creation time	2/22/2024 4:53:48 PM
---------------	----------------------

Measure (normal) [1] (Measurenormal1)

Creation time	2/22/2024 4:53:48 PM
T	25.3 °C
E	7.715 pH
tMe	01:12 min
ESTMe	4.686 pH

Titration (EQP) [1] (TitrationEQP1)

Creation time	2/22/2024 4:55:01 PM
Terminated at	Potential
EEQ1	4.574 pH ⁽⁹⁾
EHN1	6.229 pH
Q1	0.116195 mmol ⁽⁹⁾
VEQ1	1.160786 mL ⁽⁹⁾

Titration report

LabX

EST	7.723 pH
VEX	0.822964 mL
VEND	1.98375 mL
t	23:20 min
QEND	0.198573 mmol
QEX	0.082379 mmol
EPD	6.649 pH
nEQ	1

Content (R1)

Creation time	2/22/2024 5:18:26 PM ⁽⁹⁾
Result	1.9334 mmol/L ⁽⁹⁾
Decimal places	4 ⁽⁹⁾
Substance	None ⁽⁹⁾
M	1 g/mol ⁽⁹⁾
z	1 ⁽⁹⁾

Rinse (Rinse1)

Creation time	2/22/2024 5:18:27 PM
---------------	----------------------

Scope 1/1, Sample 14/17 (FTS2023 8/8 1530 rep3; 14)

Sample (S1)

Creation time	2/22/2024 5:18:38 PM
Sample start time	2/22/2024 5:18:38 PM
Sample size	57.438 g
Correction factor	1.0
Temperature	25.0 °C
Density	1.0 g/mL

Titration stand (Titrationstand1)

Creation time	2/22/2024 5:18:38 PM
---------------	----------------------

Measure (normal) [1] (Measurenormal1)

Creation time	2/22/2024 5:18:38 PM
T	25.2 °C
E	7.780 pH
tMe	01:12 min
ESTMe	4.725 pH

Titration (EQP) [1] (TitrationEQP1)

Creation time	2/22/2024 5:19:51 PM
Terminated at	Potential
EEQ1	4.583 pH ⁽⁹⁾
EHN1	6.234 pH
Q1	0.111712 mmol ⁽⁹⁾
VEQ1	1.116008 mL ⁽⁹⁾
EST	7.785 pH
VEX	0.757242 mL
VEND	1.87325 mL
t	22:21 min

Titration report

LabX

QEND 0.187512 mmol
QEX 0.075800 mmol
EPD 6.637 pH
nEQ 2 ⁽⁹⁾

Content (R1)

Creation time 2/22/2024 5:42:17 PM ⁽⁹⁾
Result 1.9449 mmol/L ⁽⁹⁾
Decimal places 4 ⁽⁹⁾
Substance None ⁽⁹⁾
M 1 g/mol ⁽⁹⁾
z 1 ⁽⁹⁾

Rinse (Rinse1)

Creation time 2/22/2024 5:42:17 PM

Scope 1/1, Sample 15/17 (FTS2023 8/8 1530 rep4; 15)

Sample (S1)

Creation time 2/22/2024 5:42:28 PM ⁽¹²⁾
Sample start time 2/22/2024 5:42:28 PM ⁽¹²⁾
Sample size 63.249 g ⁽¹²⁾
Correction factor 1.0 ⁽¹²⁾
Temperature 25.0 °C ⁽¹²⁾
Density 1.0 g/mL ⁽¹²⁾

Titration stand (Titrationstand1)

Creation time 2/22/2024 5:42:29 PM ⁽¹²⁾

Measure (normal) [1] (Measurenormal1)

Creation time 2/22/2024 5:42:29 PM ⁽¹²⁾
T 25.3 °C ⁽¹²⁾
E 7.796 pH ⁽¹²⁾
tMe 01:12 min ⁽¹²⁾
ESTMe 4.800 pH ⁽¹²⁾

Titration (EQP) [1] (TitrationEQP1)

Creation time 2/22/2024 5:43:42 PM ⁽¹²⁾
EEQ1 4.575 pH ^(9,12)
EHN1 6.232 pH ⁽¹²⁾
Q1 0.123115 mmol ^(9,12)
VEQ1 1.229921 mL ^(9,12)
nEQ 2 ^(9,12)

Measure (normal) [1] Scope 1/1, Sample 1/17

Measured Values Scope 1/1, Sample 1/17

Time [s]	Meas. value [pH]	Temp. [°C]	dE [pH]
1	4.014	21.8	

Titration report

LabX

2	4.730	22.1	0.716
3	5.056	22.6	0.326
4	5.288	23.1	0.232
5	5.470	23.5	0.182
6	5.621	23.8	0.151
7	5.755	24.1	0.134
8	5.870	24.3	0.115
9	5.979	24.5	0.109
10	6.086	24.6	0.107
11	6.185	24.7	0.099
12	6.281	24.8	0.096
13	6.370	24.8	0.089
14	6.446	24.9	0.076
15	6.526	24.9	0.080
16	6.603	24.9	0.077
17	6.677	25.0	0.074
18	6.746	25.0	0.069
19	6.813	25.0	0.067
20	6.872	25.0	0.059
21	6.929	25.0	0.057
22	6.986	25.0	0.057
23	7.031	25.0	0.045
24	7.085	25.0	0.054
25	7.135	25.0	0.050
26	7.178	25.0	0.043
27	7.220	25.0	0.042
28	7.260	25.0	0.040
29	7.297	25.0	0.037
30	7.334	25.1	0.037
31	7.368	25.1	0.034
32	7.397	25.1	0.029
33	7.427	25.1	0.030
34	7.455	25.1	0.028
35	7.480	25.1	0.025
36	7.505	25.1	0.025
37	7.528	25.1	0.023
38	7.550	25.1	0.022
39	7.568	25.1	0.018
40	7.587	25.1	0.019
41	7.606	25.1	0.019
42	7.622	25.1	0.016
43	7.637	25.1	0.015
44	7.652	25.1	0.015
45	7.664	25.1	0.012
46	7.678	25.1	0.014
47	7.689	25.1	0.011
48	7.700	25.1	0.011
49	7.711	25.1	0.011
50	7.721	25.1	0.010
51	7.730	25.0	0.009
52	7.739	25.0	0.009

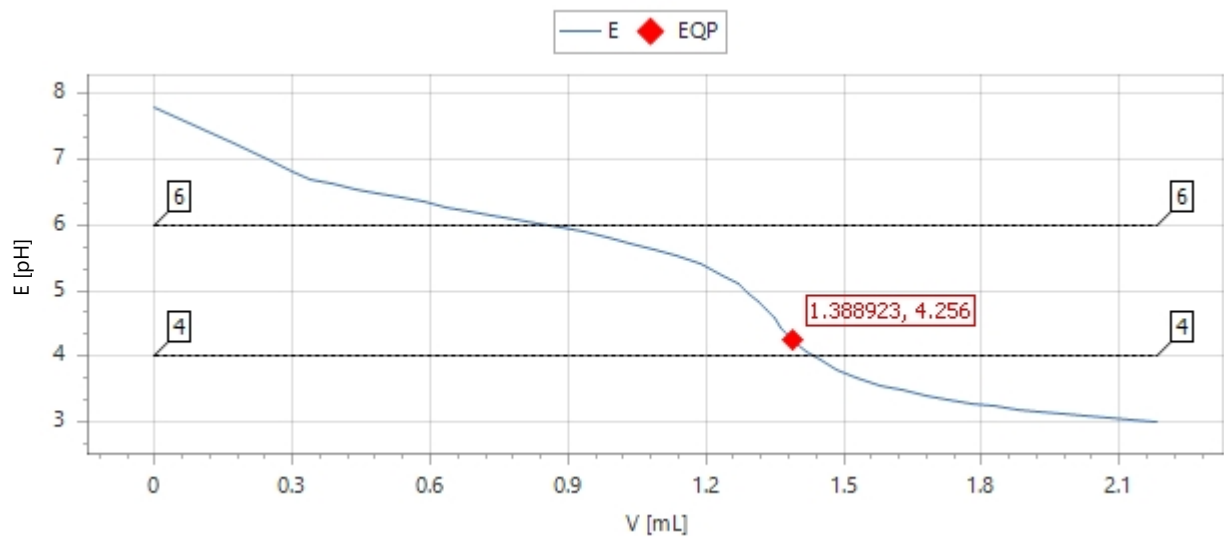
Titration report

LabX

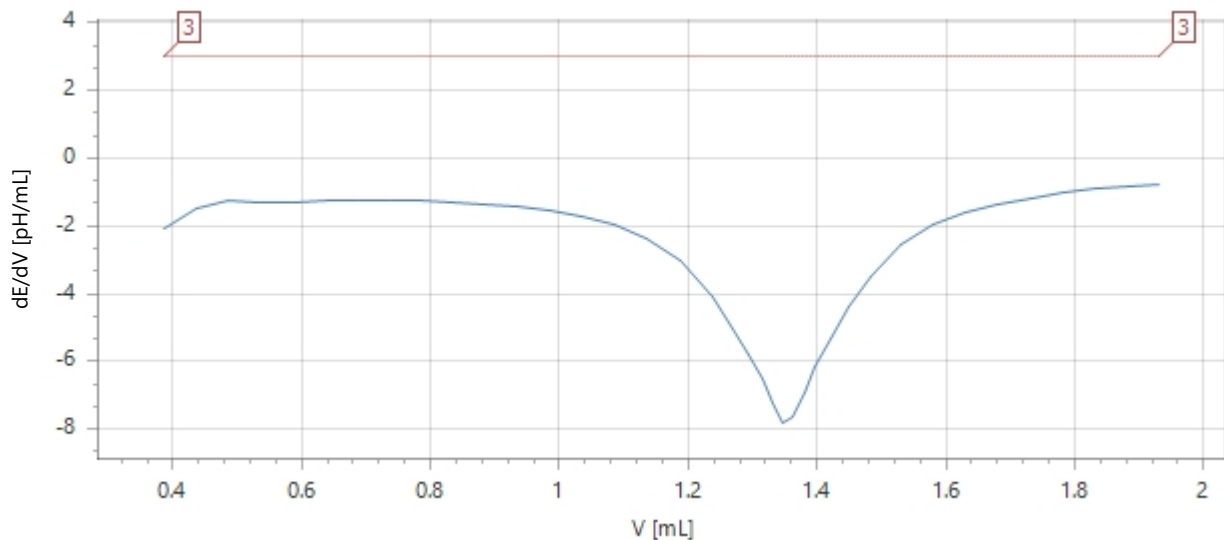
53	7.747	25.0	0.008
54	7.753	25.0	0.006
55	7.760	25.1	0.007
56	7.770	25.0	0.010
57	7.775	25.0	0.005
58	7.781	25.0	0.006
59	7.786	25.0	0.005
60	7.791	25.0	0.005

Titration (EQP) [1] Scope 1/1, Sample 1/17

E - V Curve Scope 1/1, Sample 1/17



dE/dV - V Curve Scope 1/1, Sample 1/17



Measured Values Scope 1/1, Sample 1/17

Time [s]	V [mL]	Meas. value [pH]	dV [mL]	Temp. [°C]	dE [pH]	dE/dV [pH/mL]	EQP
0	0.00000	7.801		25.0			

Titration report

LabX

28	0.17150	7.253	0.17150	25.0	-0.548	
58	0.25725	6.958	0.08575	25.0	-0.295	
89	0.30000	6.804	0.04275	25.0	-0.154	
180	0.33675	6.698	0.03675	25.0	-0.106	
210	0.38675	6.621	0.05000	25.0	-0.077	-2.08
240	0.43675	6.551	0.05000	25.0	-0.070	-1.48
270	0.48675	6.479	0.05000	25.0	-0.072	-1.27
301	0.53675	6.411	0.05000	25.0	-0.068	-1.28
331	0.58675	6.346	0.05000	25.0	-0.065	-1.29
361	0.63675	6.282	0.05000	25.0	-0.064	-1.26
391	0.68675	6.219	0.05000	25.1	-0.063	-1.25
422	0.73675	6.157	0.05000	25.1	-0.062	-1.25
452	0.78675	6.095	0.05000	25.0	-0.062	-1.27
482	0.83675	6.029	0.05000	25.0	-0.066	-1.31
512	0.88675	5.961	0.05000	25.0	-0.068	-1.37
542	0.93675	5.892	0.05000	25.1	-0.069	-1.44
573	0.98675	5.816	0.05000	25.0	-0.076	-1.55
603	1.03675	5.733	0.05000	25.0	-0.083	-1.71
633	1.08675	5.642	0.05000	25.0	-0.091	-1.98
664	1.13675	5.534	0.05000	25.0	-0.108	-2.39
694	1.18675	5.404	0.05000	25.0	-0.130	-3.04
724	1.23675	5.237	0.05000	25.0	-0.167	-4.09
754	1.26950	5.096	0.03275	25.0	-0.141	-5.01
784	1.29400	4.965	0.02450	25.1	-0.131	-5.81
815	1.31450	4.834	0.02050	25.1	-0.131	-6.53
845	1.33225	4.703	0.01775	25.1	-0.131	-7.22
875	1.34800	4.576	0.01575	25.1	-0.127	-7.80
906	1.36350	4.448	0.01550	25.1	-0.128	-7.63
936	1.37950	4.322	0.01600	25.1	-0.126	-6.91
	1.388923	4.256				EQP
966	1.39750	4.196	0.01800	25.1	-0.126	-6.16
996	1.41950	4.069	0.02200	25.1	-0.127	-5.35
1026	1.44775	3.935	0.02825	25.1	-0.134	-4.41
1057	1.48300	3.806	0.03525	25.1	-0.129	-3.47
1087	1.53025	3.674	0.04725	25.1	-0.132	-2.57
1117	1.58025	3.565	0.05000	25.1	-0.109	-1.97
1147	1.63025	3.477	0.05000	25.1	-0.088	-1.59
1178	1.68025	3.404	0.05000	25.1	-0.073	-1.34
1208	1.73025	3.341	0.05000	25.1	-0.063	-1.16
1238	1.78025	3.286	0.05000	25.1	-0.055	-1.02
1268	1.83025	3.237	0.05000	25.1	-0.049	-0.92
1299	1.88025	3.193	0.05000	25.1	-0.044	-0.83
1329	1.93025	3.153	0.05000	25.1	-0.040	-0.76
1359	1.98025	3.117	0.05000	25.1	-0.036	
1389	2.03025	3.083	0.05000	25.1	-0.034	
1420	2.08025	3.052	0.05000	25.1	-0.031	
1450	2.13025	3.020	0.05000	25.1	-0.032	
1480	2.18025	2.995	0.05000	25.1	-0.025	

Measure (normal) [1] Scope 1/1, Sample 2/17

Titration report

LabX

Measured Values Scope 1/1, Sample 2/17

Time [s]	Meas. value [pH]	Temp. [°C]	dE [pH]
1	4.644	23.6	
2	5.024	23.7	0.380
3	5.296	24.0	0.272
4	5.512	24.2	0.216
5	5.693	24.4	0.181
6	5.852	24.6	0.159
7	5.998	24.7	0.146
8	6.117	24.8	0.119
9	6.245	24.9	0.128
10	6.364	25.0	0.119
11	6.470	25.0	0.106
12	6.574	25.1	0.104
13	6.669	25.1	0.095
14	6.760	25.1	0.091
15	6.837	25.1	0.077
16	6.914	25.1	0.077
17	6.986	25.2	0.072
18	7.049	25.2	0.063
19	7.115	25.2	0.066
20	7.175	25.2	0.060
21	7.230	25.2	0.055
22	7.276	25.2	0.046
23	7.323	25.2	0.047
24	7.367	25.2	0.044
25	7.407	25.2	0.040
26	7.444	25.2	0.037
27	7.479	25.2	0.035
28	7.511	25.2	0.032
29	7.538	25.2	0.027
30	7.566	25.2	0.028
31	7.592	25.2	0.026
32	7.615	25.2	0.023
33	7.637	25.2	0.022
34	7.657	25.2	0.020
35	7.677	25.2	0.020
36	7.692	25.2	0.015
37	7.708	25.2	0.016
38	7.723	25.2	0.015
39	7.736	25.2	0.013
40	7.749	25.2	0.013
41	7.761	25.2	0.012
42	7.771	25.2	0.010
43	7.780	25.2	0.009
44	7.789	25.2	0.009
45	7.798	25.2	0.009
46	7.806	25.2	0.008
47	7.813	25.2	0.007

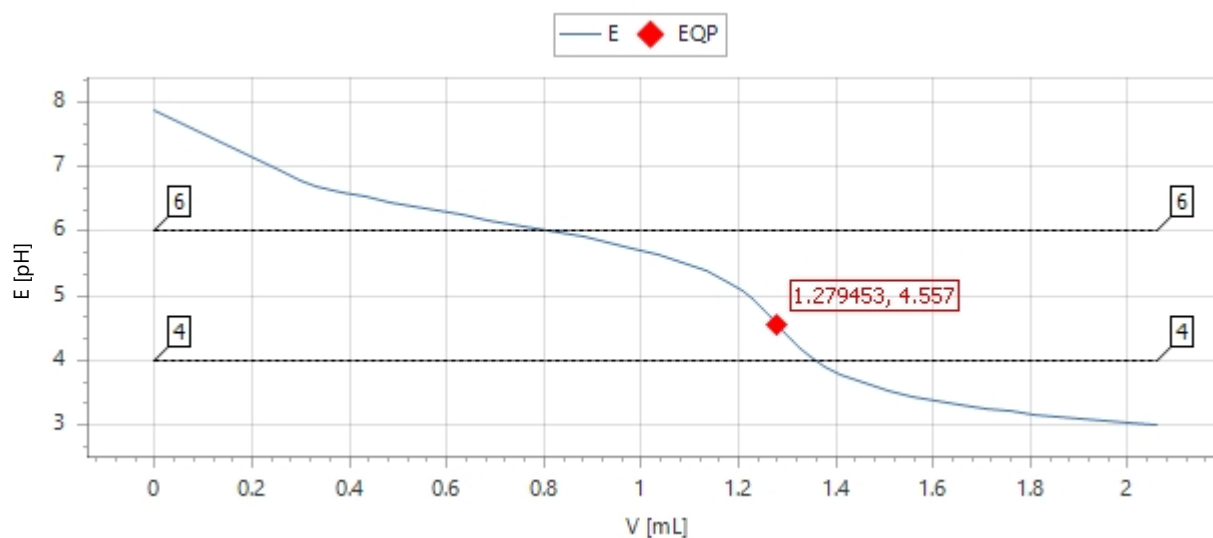
Titration report

LabX

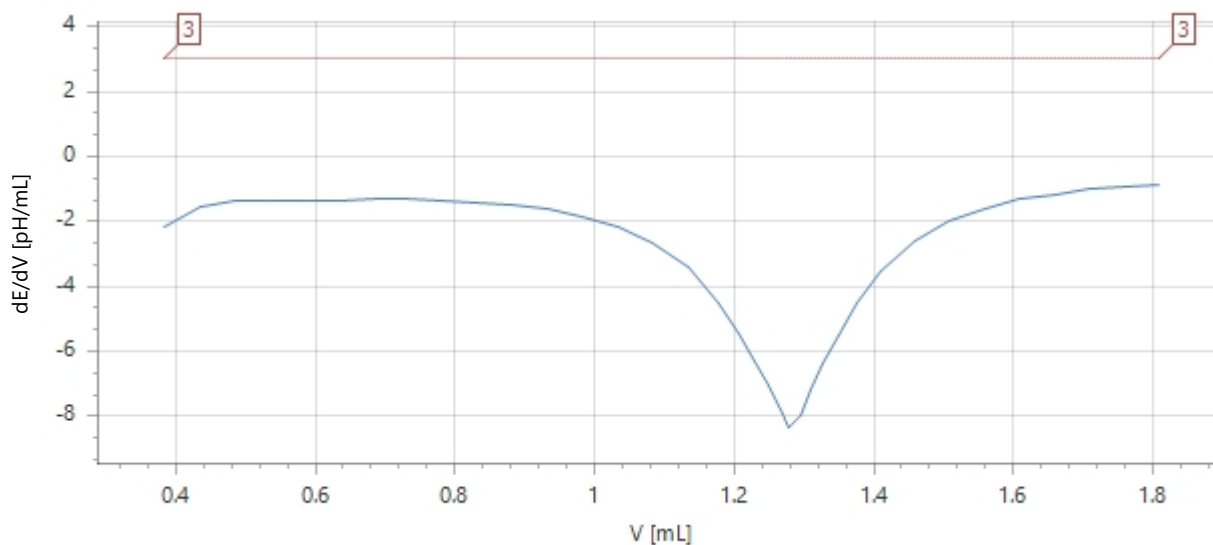
48	7.820	25.2	0.007
49	7.826	25.2	0.006
50	7.832	25.2	0.006
51	7.837	25.2	0.005
52	7.842	25.2	0.005
53	7.847	25.2	0.005
54	7.851	25.2	0.004
55	7.856	25.2	0.005
56	7.859	25.2	0.003
57	7.863	25.2	0.004
58	7.866	25.2	0.003
59	7.868	25.2	0.002
60	7.871	25.2	0.003

Titration (EQP) [1] Scope 1/1, Sample 2/17

E - V Curve Scope 1/1, Sample 2/17



dE/dV - V Curve Scope 1/1, Sample 2/17



Titration report

LabX

Measured Values Scope 1/1, Sample 2/17

Time [s]	V [mL]	Meas. value [pH]	dV [mL]	Temp. [°C]	dE [pH]	dE/dV [pH/mL]	EQP
0	0.00000	7.877		25.2			
29	0.17150	7.260	0.17150	25.2	-0.617		
59	0.25725	6.951	0.08575	25.2	-0.309		
90	0.30000	6.789	0.04275	25.2	-0.162		
181	0.33375	6.682	0.03375	25.2	-0.107		
211	0.38375	6.610	0.05000	25.2	-0.072	-2.15	
241	0.43375	6.532	0.05000	25.2	-0.078	-1.55	
271	0.48375	6.457	0.05000	25.2	-0.075	-1.34	
302	0.53375	6.384	0.05000	25.2	-0.073	-1.36	
332	0.58375	6.316	0.05000	25.1	-0.068	-1.38	
362	0.63375	6.249	0.05000	25.1	-0.067	-1.34	
392	0.68375	6.178	0.05000	25.1	-0.071	-1.32	
423	0.73375	6.116	0.05000	25.1	-0.062	-1.32	
453	0.78375	6.048	0.05000	25.1	-0.068	-1.36	
483	0.83375	5.977	0.05000	25.1	-0.071	-1.42	
513	0.88375	5.908	0.05000	25.1	-0.069	-1.51	
544	0.93375	5.823	0.05000	25.1	-0.085	-1.64	
574	0.98375	5.736	0.05000	25.1	-0.087	-1.84	
604	1.03375	5.638	0.05000	25.1	-0.098	-2.15	
634	1.08375	5.522	0.05000	25.1	-0.116	-2.63	
665	1.13375	5.379	0.05000	25.1	-0.143	-3.41	
695	1.17875	5.211	0.04500	25.1	-0.168	-4.54	
725	1.20675	5.081	0.02800	25.1	-0.130	-5.45	
755	1.23000	4.944	0.02325	25.1	-0.137	-6.30	
786	1.24825	4.819	0.01825	25.1	-0.125	-7.02	
816	1.26525	4.682	0.01700	25.1	-0.137	-7.79	
	1.279453 ⁽⁹⁾	4.557 ⁽⁹⁾					EQP
846	1.27950	4.557	0.01425	25.1	-0.125	-8.32	
876	1.29400	4.432	0.01450	25.1	-0.125	-7.99	
906	1.30975	4.302	0.01575	25.1	-0.130	-7.17	
937	1.32700	4.179	0.01725	25.1	-0.123	-6.39	
967	1.34875	4.049	0.02175	25.1	-0.130	-5.51	
997	1.37600	3.918	0.02725	25.1	-0.131	-4.53	
1028	1.41125	3.787	0.03525	25.1	-0.131	-3.53	
1058	1.45825	3.654	0.04700	25.0	-0.133	-2.60	
1088	1.50825	3.544	0.05000	25.0	-0.110	-1.98	
1118	1.55825	3.456	0.05000	25.0	-0.088	-1.59	
1148	1.60825	3.383	0.05000	25.0	-0.073	-1.33	
1179	1.65825	3.320	0.05000	25.0	-0.063	-1.15	
1209	1.70825	3.265	0.05000	25.0	-0.055	-1.02	
1239	1.75825	3.216	0.05000	25.0	-0.049	-0.92	
1269	1.80825	3.172	0.05000	25.0	-0.044	-0.84	
1300	1.85825	3.132	0.05000	25.0	-0.040		
1330	1.90825	3.095	0.05000	25.0	-0.037		
1360	1.95825	3.061	0.05000	25.0	-0.034		
1390	2.00825	3.029	0.05000	25.0	-0.032		
1421	2.05825	3.000	0.05000	25.0	-0.029		

Titration report

LabX

Measure (normal) [1] Scope 1/1, Sample 3/17

Measured Values Scope 1/1, Sample 3/17

Time [s]	Meas. value [pH]	Temp. [°C]	dE [pH]
1	4.593	23.5	
2	4.897	23.6	0.304
3	5.147	23.9	0.250
4	5.331	24.1	0.184
5	5.478	24.3	0.147
6	5.600	24.5	0.122
7	5.702	24.7	0.102
8	5.784	24.8	0.082
9	5.867	24.9	0.083
10	5.945	24.9	0.078
11	6.015	25.0	0.070
12	6.081	25.0	0.066
13	6.143	25.1	0.062
14	6.203	25.1	0.060
15	6.253	25.1	0.050
16	6.306	25.1	0.053
17	6.357	25.1	0.051
18	6.406	25.2	0.049
19	6.454	25.2	0.048
20	6.501	25.2	0.047
21	6.545	25.2	0.044
22	6.582	25.2	0.037
23	6.623	25.2	0.041
24	6.661	25.2	0.038
25	6.699	25.2	0.038
26	6.736	25.2	0.037
27	6.771	25.2	0.035
28	6.806	25.2	0.035
29	6.837	25.2	0.031
30	6.870	25.2	0.033
31	6.899	25.2	0.029
32	6.925	25.2	0.026
33	6.954	25.2	0.029
34	6.981	25.2	0.027
35	7.007	25.2	0.026
36	7.024	25.2	0.017
37	7.050	25.2	0.026
38	7.076	25.2	0.026
39	7.098	25.2	0.022
40	7.121	25.2	0.023
41	7.142	25.2	0.021
42	7.162	25.2	0.020
43	7.179	25.2	0.017
44	7.197	25.2	0.018
45	7.215	25.2	0.018

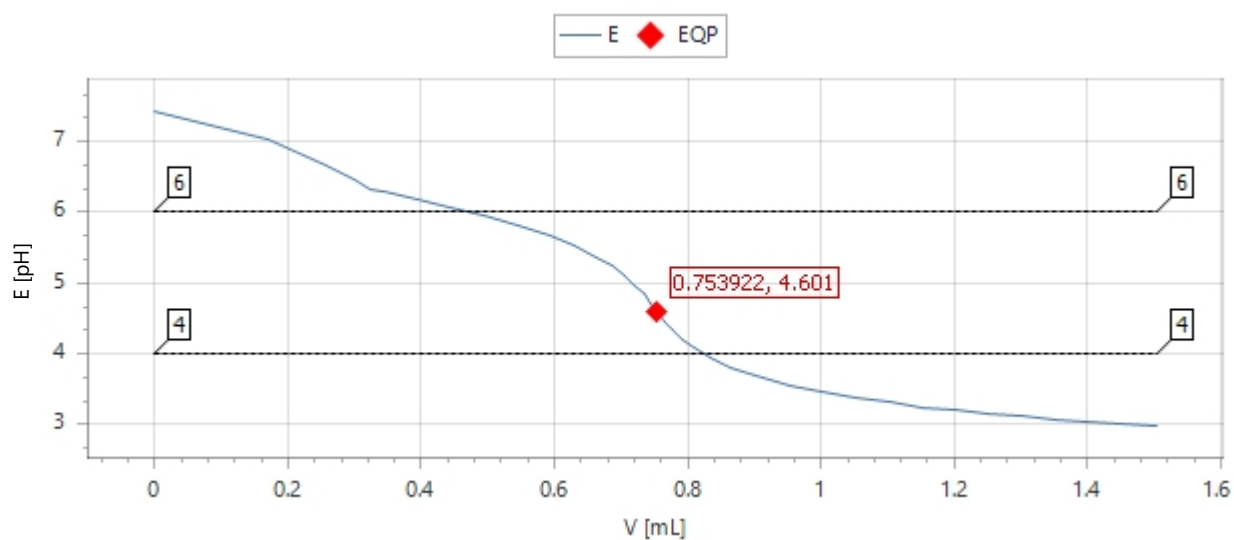
Titration report

LabX

46	7.232	25.2	0.017
47	7.248	25.2	0.016
48	7.264	25.2	0.016
49	7.280	25.2	0.016
50	7.292	25.2	0.012
51	7.306	25.2	0.014
52	7.320	25.2	0.014
53	7.334	25.2	0.014
54	7.347	25.2	0.013
55	7.359	25.2	0.012
56	7.371	25.2	0.012
57	7.381	25.2	0.010
58	7.392	25.2	0.011
59	7.402	25.2	0.010
60	7.412	25.2	0.010

Titration (EQP) [1] Scope 1/1, Sample 3/17

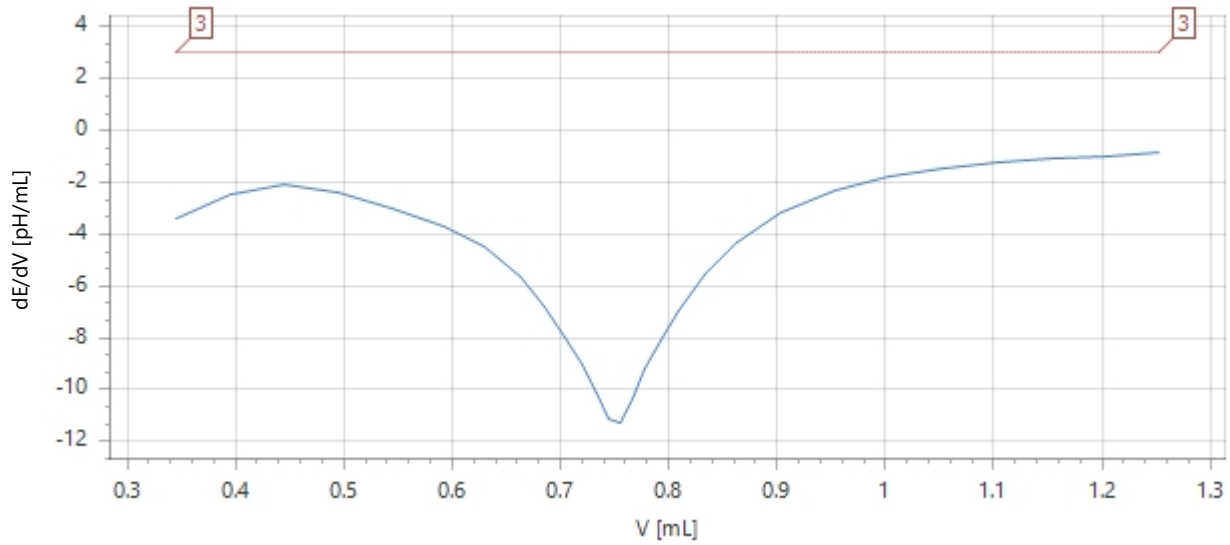
E - V Curve Scope 1/1, Sample 3/17



dE/dV - V Curve Scope 1/1, Sample 3/17

Titration report

LabX



Measured Values Scope 1/1, Sample 3/17

Time [s]	V [mL]	Meas. value [pH]	dV [mL]	Temp. [°C]	dE [pH]	dE/dV [pH/mL]	EQP
0	0.00000	7.435		25.2			
28	0.17150	7.021	0.17150	25.2	-0.414		
58	0.25725	6.671	0.08575	25.2	-0.350		
89	0.30000	6.456	0.04275	25.2	-0.215		
180	0.32200	6.335	0.02200	25.2	-0.121		
210	0.34475	6.291	0.02275	25.1	-0.044	-3.36	
240	0.39475	6.190	0.05000	25.1	-0.101	-2.48	
270	0.44475	6.074	0.05000	25.1	-0.116	-2.11	
300	0.49475	5.951	0.05000	25.1	-0.123	-2.41	
331	0.54475	5.813	0.05000	25.1	-0.138	-2.97	
361	0.59275	5.659	0.04800	25.1	-0.154	-3.67	
391	0.62925	5.527	0.03650	25.1	-0.132	-4.46	
422	0.66275	5.372	0.03350	25.1	-0.155	-5.63	
452	0.68575	5.240	0.02300	25.1	-0.132	-6.76	
482	0.70450	5.109	0.01875	25.1	-0.131	-7.92	
512	0.72050	4.973	0.01600	25.1	-0.136	-9.07	
542	0.73350	4.838	0.01300	25.1	-0.135	-10.10	
573	0.74425	4.717	0.01075	25.1	-0.121	-11.10	
	0.753922	4.601					EQP
603	0.75550	4.582	0.01125	25.1	-0.135	-11.25	
633	0.76600	4.458	0.01050	25.1	-0.124	-10.35	
663	0.77775	4.335	0.01175	25.1	-0.123	-9.22	
694	0.79200	4.207	0.01425	25.1	-0.128	-8.13	
724	0.80975	4.078	0.01775	25.1	-0.129	-6.92	
754	0.83275	3.946	0.02300	25.1	-0.132	-5.57	
784	0.86300	3.813	0.03025	25.1	-0.133	-4.28	
815	0.90300	3.681	0.04000	25.1	-0.132	-3.14	
845	0.95300	3.556	0.05000	25.1	-0.125	-2.27	
875	1.00300	3.458	0.05000	25.1	-0.098	-1.76	
905	1.05300	3.379	0.05000	25.1	-0.079	-1.45	
936	1.10300	3.310	0.05000	25.1	-0.069	-1.24	

Titration report

LabX

966	1.15300	3.252	0.05000	25.1	-0.058	-1.08
996	1.20300	3.200	0.05000	25.1	-0.052	-0.97
1026	1.25300	3.154	0.05000	25.1	-0.046	-0.87
1057	1.30300	3.112	0.05000	25.1	-0.042	
1087	1.35300	3.074	0.05000	25.1	-0.038	
1117	1.40300	3.039	0.05000	25.1	-0.035	
1147	1.45300	3.007	0.05000	25.1	-0.032	
1178	1.50300	2.976	0.05000	25.1	-0.031	

Measure (normal) [1] Scope 1/1, Sample 4/17

Measured Values Scope 1/1, Sample 4/17

Time [s]	Meas. value [pH]	Temp. [°C]	dE [pH]
1	4.511	23.5	
2	4.876	23.7	0.365
3	5.133	23.9	0.257
4	5.328	24.2	0.195
5	5.481	24.4	0.153
6	5.608	24.6	0.127
7	5.717	24.7	0.109
8	5.804	24.8	0.087
9	5.892	24.9	0.088
10	5.974	25.0	0.082
11	6.049	25.1	0.075
12	6.120	25.1	0.071
13	6.186	25.1	0.066
14	6.250	25.2	0.064
15	6.305	25.2	0.055
16	6.363	25.2	0.058
17	6.418	25.2	0.055
18	6.470	25.2	0.052
19	6.520	25.2	0.050
20	6.569	25.2	0.049
21	6.615	25.2	0.046
22	6.655	25.2	0.040
23	6.698	25.2	0.043
24	6.738	25.2	0.040
25	6.778	25.2	0.040
26	6.817	25.2	0.039
27	6.855	25.2	0.038
28	6.890	25.2	0.035
29	6.917	25.2	0.027
30	6.950	25.2	0.033
31	6.980	25.2	0.030
32	7.010	25.2	0.030
33	7.031	25.2	0.021
34	7.064	25.2	0.033
35	7.091	25.2	0.027
36	7.115	25.2	0.024

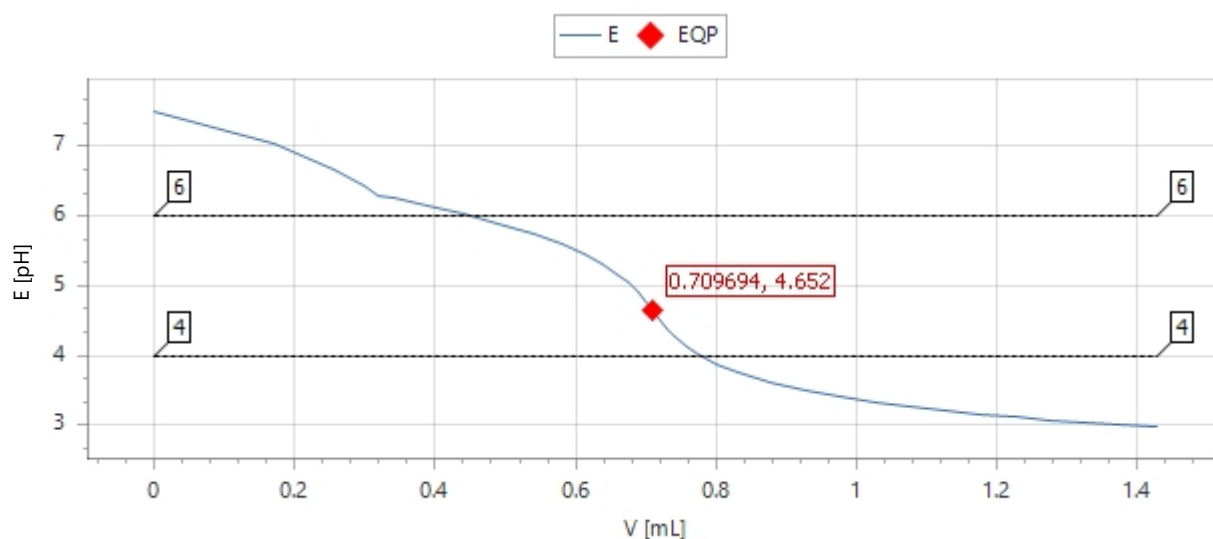
Titration report

LabX

37	7.141	25.2	0.026
38	7.164	25.2	0.023
39	7.186	25.2	0.022
40	7.207	25.2	0.021
41	7.228	25.2	0.021
42	7.245	25.2	0.017
43	7.264	25.2	0.019
44	7.283	25.2	0.019
45	7.301	25.2	0.018
46	7.317	25.2	0.016
47	7.333	25.2	0.016
48	7.348	25.2	0.015
49	7.361	25.2	0.013
50	7.375	25.2	0.014
51	7.389	25.2	0.014
52	7.401	25.2	0.012
53	7.414	25.2	0.013
54	7.426	25.2	0.012
55	7.438	25.2	0.012
56	7.447	25.2	0.009
57	7.457	25.2	0.010
58	7.467	25.2	0.010
59	7.477	25.2	0.010
60	7.486	25.2	0.009

Titration (EQP) [1] Scope 1/1, Sample 4/17

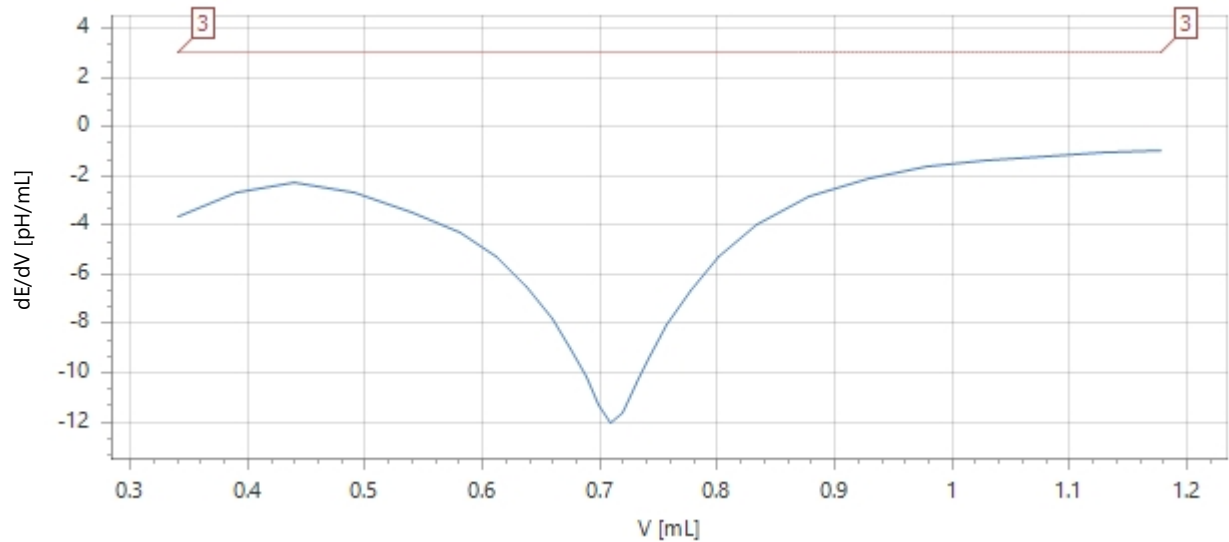
E - V Curve Scope 1/1, Sample 4/17



dE/dV - V Curve Scope 1/1, Sample 4/17

Titration report

LabX



Measured Values Scope 1/1, Sample 4/17

Time [s]	V [mL]	Meas. value [pH]	dV [mL]	Temp. [°C]	dE [pH]	dE/dV [pH/mL]	EQP
0	0.00000	7.507		25.2			
30	0.17150	7.027	0.17150	25.2	-0.480		
60	0.25725	6.656	0.08575	25.2	-0.371		
90	0.30000	6.423	0.04275	25.2	-0.233		
182	0.32000	6.302	0.02000	25.2	-0.121		
212	0.34025	6.263	0.02025	25.2	-0.039	-3.63	
242	0.39025	6.151	0.05000	25.2	-0.112	-2.68	
272	0.44025	6.025	0.05000	25.2	-0.126	-2.24	
303	0.49025	5.889	0.05000	25.2	-0.136	-2.65	
333	0.54025	5.739	0.05000	25.2	-0.150	-3.45	
363	0.58125	5.589	0.04100	25.2	-0.150	-4.33	
393	0.61150	5.458	0.03025	25.2	-0.131	-5.28	
424	0.63800	5.316	0.02650	25.1	-0.142	-6.52	
454	0.65875	5.176	0.02075	25.1	-0.140	-7.82	
484	0.67475	5.046	0.01600	25.1	-0.130	-9.00	
514	0.68850	4.907	0.01375	25.1	-0.139	-10.19	
545	0.69925	4.783	0.01075	25.1	-0.124	-11.27	
	0.709694	4.652					EQP
575	0.70975	4.651	0.01050	25.1	-0.132	-12.00	
605	0.71950	4.529	0.00975	25.1	-0.122	-11.59	
635	0.73050	4.397	0.01100	25.1	-0.132	-10.40	
666	0.74250	4.272	0.01200	25.1	-0.125	-9.23	
696	0.75725	4.145	0.01475	25.1	-0.127	-8.04	
726	0.77650	4.011	0.01925	25.1	-0.134	-6.65	
756	0.80050	3.882	0.02400	25.1	-0.129	-5.29	
786	0.83300	3.751	0.03250	25.1	-0.131	-3.99	
817	0.87825	3.613	0.04525	25.1	-0.138	-2.83	
847	0.92825	3.498	0.05000	25.1	-0.115	-2.08	
877	0.97825	3.406	0.05000	25.1	-0.092	-1.64	
908	1.02825	3.331	0.05000	25.1	-0.075	-1.37	
938	1.07825	3.269	0.05000	25.1	-0.062	-1.18	

Titration report

LabX

968	1.12825	3.211	0.05000	25.1	-0.058	-1.05
998	1.17825	3.160	0.05000	25.1	-0.051	-0.94
1028	1.22825	3.115	0.05000	25.1	-0.045	
1059	1.27825	3.075	0.05000	25.1	-0.040	
1089	1.32825	3.038	0.05000	25.1	-0.037	
1119	1.37825	3.004	0.05000	25.1	-0.034	
1149	1.42825	2.972	0.05000	25.1	-0.032	

Measure (normal) [1] Scope 1/1, Sample 5/17

Measured Values Scope 1/1, Sample 5/17

Time [s]	Meas. value [pH]	Temp. [°C]	dE [pH]
1	4.629	23.7	
2	4.874	23.8	0.245
3	5.111	24.1	0.237
4	5.290	24.3	0.179
5	5.429	24.5	0.139
6	5.540	24.7	0.111
7	5.637	24.8	0.097
8	5.714	24.9	0.077
9	5.790	25.0	0.076
10	5.846	25.0	0.056
11	5.927	25.1	0.081
12	5.993	25.1	0.066
13	6.043	25.1	0.050
14	6.087	25.2	0.044
15	6.135	25.2	0.048
16	6.186	25.2	0.051
17	6.230	25.2	0.044
18	6.266	25.2	0.036
19	6.316	25.2	0.050
20	6.355	25.2	0.039
21	6.389	25.2	0.034
22	6.426	25.2	0.037
23	6.462	25.2	0.036
24	6.496	25.2	0.034
25	6.529	25.2	0.033
26	6.559	25.2	0.030
27	6.589	25.2	0.030
28	6.622	25.2	0.033
29	6.647	25.2	0.025
30	6.674	25.2	0.027
31	6.701	25.2	0.027
32	6.727	25.2	0.026
33	6.750	25.2	0.023
34	6.775	25.2	0.025
35	6.800	25.2	0.025
36	6.824	25.2	0.024
37	6.848	25.2	0.024

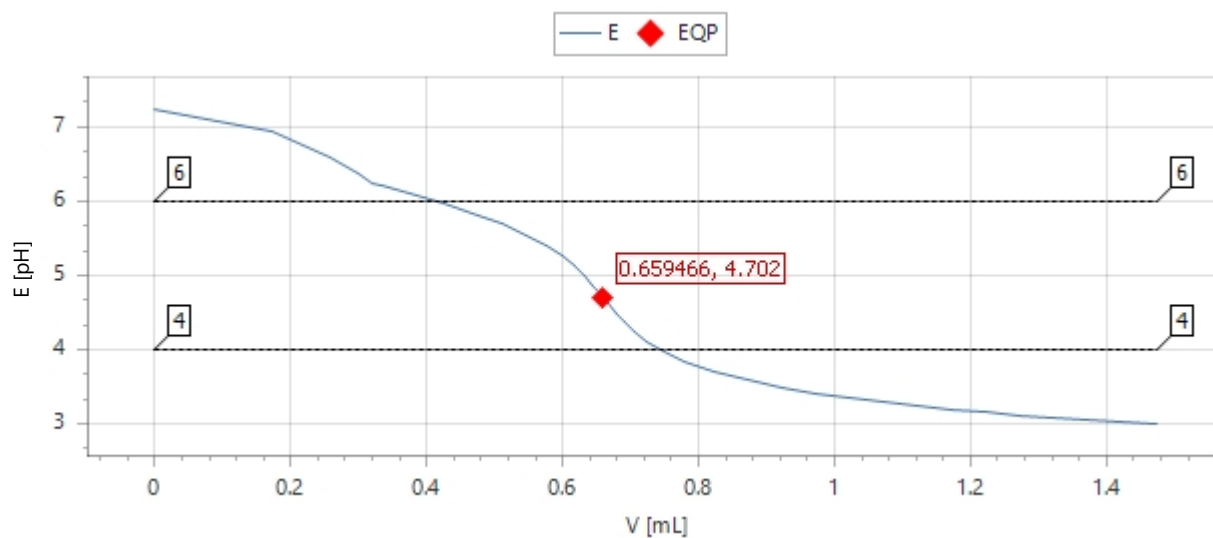
Titration report

LabX

38	6.873	25.2	0.025
39	6.894	25.2	0.021
40	6.911	25.2	0.017
41	6.930	25.2	0.019
42	6.951	25.2	0.021
43	6.969	25.2	0.018
44	6.988	25.2	0.019
45	7.005	25.2	0.017
46	7.018	25.2	0.013
47	7.029	25.2	0.011
48	7.050	25.2	0.021
49	7.067	25.2	0.017
50	7.086	25.2	0.019
51	7.100	25.2	0.014
52	7.113	25.2	0.013
53	7.128	25.2	0.015
54	7.142	25.2	0.014
55	7.156	25.2	0.014
56	7.169	25.2	0.013
57	7.180	25.2	0.011
58	7.197	25.2	0.017
59	7.206	25.2	0.009
60	7.216	25.2	0.010

Titration (EQP) [1] Scope 1/1, Sample 5/17

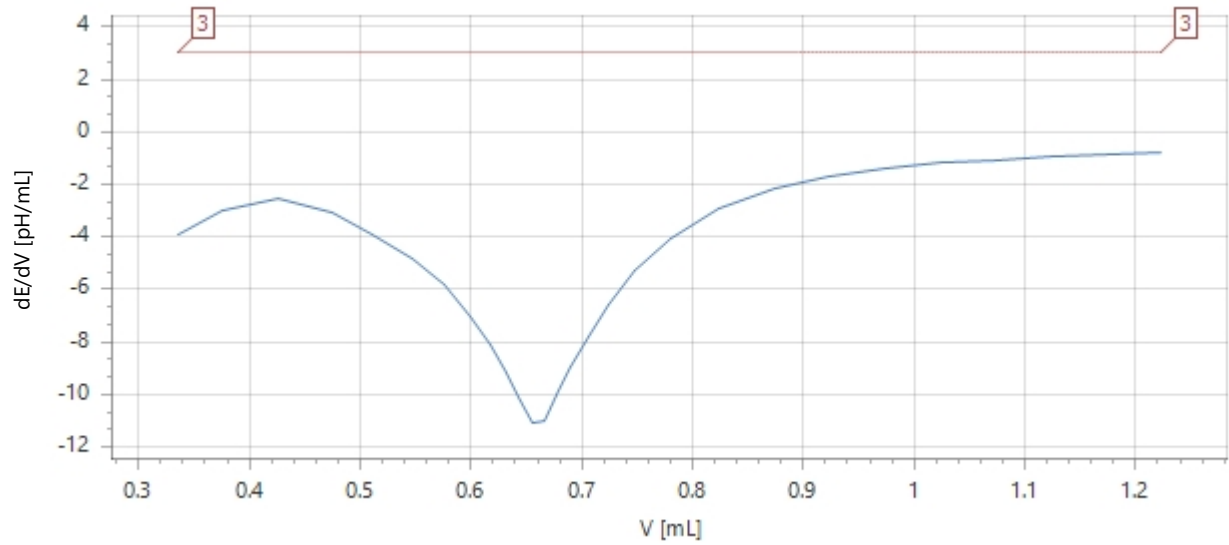
E - V Curve Scope 1/1, Sample 5/17



dE/dV - V Curve Scope 1/1, Sample 5/17

Titration report

LabX



Measured Values Scope 1/1, Sample 5/17

Time [s]	V [mL]	Meas. value [pH]	dV [mL]	Temp. [°C]	dE [pH]	dE/dV [pH/mL]	EQP
0	0.00000	7.244		25.2			
28	0.17150	6.929	0.17150	25.2	-0.315		
58	0.25725	6.589	0.08575	25.2	-0.340		
88	0.30000	6.364	0.04275	25.2	-0.225		
180	0.31925	6.235	0.01925	25.2	-0.129		
210	0.33525	6.203	0.01600	25.2	-0.032	-3.88	
240	0.37525	6.114	0.04000	25.2	-0.089	-2.98	
270	0.42525	5.980	0.05000	25.2	-0.134	-2.53	
301	0.47525	5.821	0.05000	25.2	-0.159	-3.09	
331	0.51175	5.695	0.03650	25.2	-0.126	-3.93	
361	0.54750	5.548	0.03575	25.2	-0.147	-4.85	
391	0.57550	5.411	0.02800	25.1	-0.137	-5.85	
422	0.59900	5.267	0.02350	25.1	-0.144	-7.01	
452	0.61675	5.138	0.01775	25.1	-0.129	-8.12	
482	0.63225	5.002	0.01550	25.1	-0.136	-9.20	
512	0.64500	4.870	0.01275	25.1	-0.132	-10.25	
543	0.65600	4.743	0.01100	25.1	-0.127	-11.05	
	0.659466	4.702					EQP
573	0.66650	4.620	0.01050	25.1	-0.123	-11.00	
603	0.67800	4.487	0.01150	25.1	-0.133	-10.01	
633	0.69000	4.367	0.01200	25.1	-0.120	-8.93	
664	0.70550	4.234	0.01550	25.1	-0.133	-7.82	
694	0.72425	4.105	0.01875	25.1	-0.129	-6.58	
724	0.74850	3.973	0.02425	25.1	-0.132	-5.27	
754	0.78025	3.841	0.03175	25.1	-0.132	-4.03	
785	0.82325	3.707	0.04300	25.1	-0.134	-2.93	
815	0.87325	3.589	0.05000	25.1	-0.118	-2.15	
845	0.92325	3.495	0.05000	25.1	-0.094	-1.69	
875	0.97325	3.417	0.05000	25.1	-0.078	-1.40	
906	1.02325	3.352	0.05000	25.1	-0.065	-1.20	
936	1.07325	3.294	0.05000	25.1	-0.058	-1.06	

Titration report

LabX

966	1.12325	3.244	0.05000	25.1	-0.050	-0.95
996	1.17325	3.198	0.05000	25.1	-0.046	-0.86
1026	1.22325	3.157	0.05000	25.1	-0.041	-0.78
1057	1.27325	3.120	0.05000	25.1	-0.037	
1087	1.32325	3.085	0.05000	25.1	-0.035	
1117	1.37325	3.053	0.05000	25.1	-0.032	
1148	1.42325	3.024	0.05000	25.1	-0.029	
1178	1.47325	2.996	0.05000	25.1	-0.028	

Measure (normal) [1] Scope 1/1, Sample 6/17

Measured Values Scope 1/1, Sample 6/17

Time [s]	Meas. value [pH]	Temp. [°C]	dE [pH]
1	4.604	23.5	
2	4.905	23.7	0.301
3	5.135	23.9	0.230
4	5.307	24.2	0.172
5	5.442	24.4	0.135
6	5.555	24.6	0.113
7	5.653	24.8	0.098
8	5.729	24.9	0.076
9	5.807	25.0	0.078
10	5.878	25.0	0.071
11	5.944	25.1	0.066
12	6.005	25.1	0.061
13	6.062	25.2	0.057
14	6.116	25.2	0.054
15	6.162	25.2	0.046
16	6.210	25.2	0.048
17	6.256	25.2	0.046
18	6.300	25.3	0.044
19	6.342	25.3	0.042
20	6.383	25.3	0.041
21	6.422	25.3	0.039
22	6.455	25.3	0.033
23	6.491	25.3	0.036
24	6.525	25.3	0.034
25	6.559	25.3	0.034
26	6.592	25.3	0.033
27	6.624	25.3	0.032
28	6.655	25.3	0.031
29	6.683	25.3	0.028
30	6.713	25.3	0.030
31	6.742	25.3	0.029
32	6.769	25.3	0.027
33	6.796	25.3	0.027
34	6.822	25.3	0.026
35	6.849	25.3	0.027
36	6.871	25.3	0.022

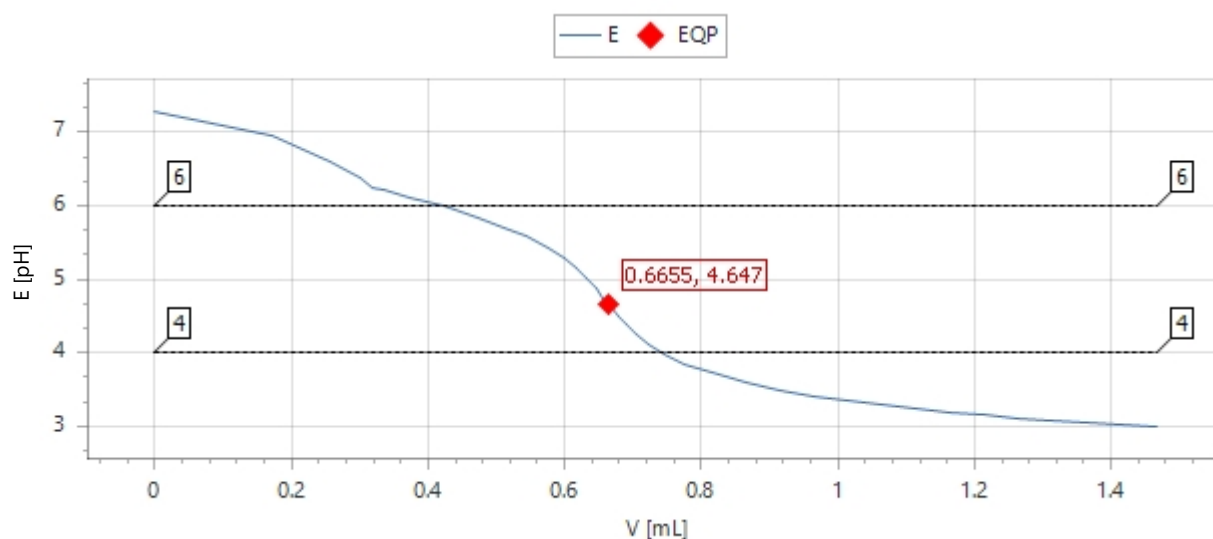
Titration report

LabX

37	6.894	25.3	0.023
38	6.914	25.3	0.020
39	6.936	25.3	0.022
40	6.957	25.3	0.021
41	6.977	25.3	0.020
42	6.998	25.3	0.021
43	7.015	25.3	0.017
44	7.026	25.3	0.011
45	7.048	25.3	0.022
46	7.069	25.3	0.021
47	7.087	25.3	0.018
48	7.104	25.3	0.017
49	7.120	25.3	0.016
50	7.135	25.3	0.015
51	7.150	25.3	0.015
52	7.165	25.3	0.015
53	7.179	25.3	0.014
54	7.192	25.3	0.013
55	7.205	25.3	0.013
56	7.218	25.3	0.013
57	7.229	25.3	0.011
58	7.241	25.3	0.012
59	7.253	25.3	0.012
60	7.263	25.3	0.010

Titration (EQP) [1] Scope 1/1, Sample 6/17

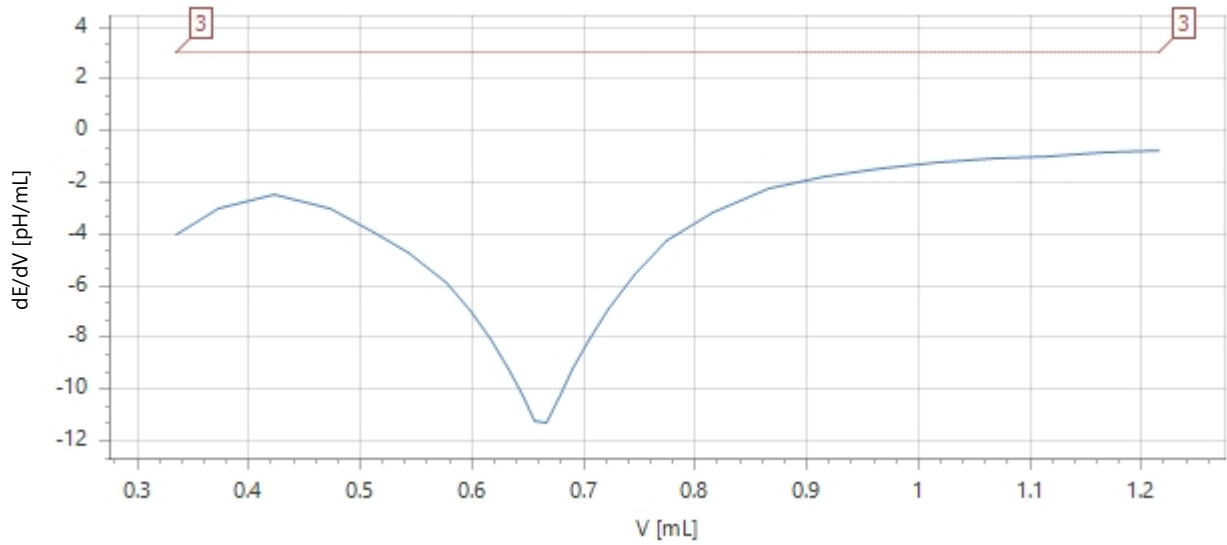
E - V Curve Scope 1/1, Sample 6/17



dE/dV - V Curve Scope 1/1, Sample 6/17

Titration report

LabX



Measured Values Scope 1/1, Sample 6/17

Time [s]	V [mL]	Meas. value [pH]	dV [mL]	Temp. [°C]	dE [pH]	dE/dV [pH/mL]	EQP
0	0.00000	7.290		25.3			
28	0.17150	6.966	0.17150	25.3	-0.324		
58	0.25725	6.605	0.08575	25.3	-0.361		
88	0.30000	6.371	0.04275	25.2	-0.234		
179	0.31900	6.239	0.01900	25.2	-0.132		
209	0.33450	6.206	0.01550	25.2	-0.033	-4.00	
239	0.37325	6.122	0.03875	25.2	-0.084	-2.98	
270	0.42325	5.994	0.05000	25.2	-0.128	-2.48	
300	0.47325	5.839	0.05000	25.2	-0.155	-3.04	
330	0.51175	5.698	0.03850	25.2	-0.141	-3.91	
360	0.54300	5.578	0.03125	25.2	-0.120	-4.68	
390	0.57650	5.415	0.03350	25.2	-0.163	-5.86	
421	0.59875	5.282	0.02225	25.2	-0.133	-6.93	
451	0.61700	5.150	0.01825	25.2	-0.132	-8.05	
481	0.63275	5.014	0.01575	25.2	-0.136	-9.23	
511	0.64575	4.879	0.01300	25.2	-0.135	-10.22	
542	0.65675	4.751	0.01100	25.2	-0.128	-11.19	
	0.665500	4.647					EQP
572	0.66700	4.629	0.01025	25.2	-0.122	-11.27	
602	0.67825	4.496	0.01125	25.2	-0.133	-10.36	
632	0.68975	4.376	0.01150	25.2	-0.120	-9.24	
663	0.70450	4.244	0.01475	25.2	-0.132	-8.12	
693	0.72225	4.115	0.01775	25.2	-0.129	-6.89	
723	0.74525	3.983	0.02300	25.2	-0.132	-5.56	
753	0.77525	3.852	0.03000	25.2	-0.131	-4.28	
784	0.81600	3.718	0.04075	25.2	-0.134	-3.13	
814	0.86600	3.593	0.05000	25.2	-0.125	-2.26	
844	0.91600	3.496	0.05000	25.2	-0.097	-1.76	
874	0.96600	3.416	0.05000	25.2	-0.080	-1.44	
905	1.01600	3.348	0.05000	25.2	-0.068	-1.23	
935	1.06600	3.290	0.05000	25.2	-0.058	-1.08	

Titration report

LabX

965	1.11600	3.238	0.05000	25.2	-0.052	-0.97
995	1.16600	3.191	0.05000	25.2	-0.047	-0.87
1026	1.21600	3.150	0.05000	25.2	-0.041	-0.79
1056	1.26600	3.112	0.05000	25.2	-0.038	
1086	1.31600	3.077	0.05000	25.2	-0.035	
1116	1.36600	3.045	0.05000	25.2	-0.032	
1146	1.41600	3.015	0.05000	25.2	-0.030	
1177	1.46600	2.987	0.05000	25.2	-0.028	

Measure (normal) [1] Scope 1/1, Sample 7/17

Measured Values Scope 1/1, Sample 7/17

Time [s]	Meas. value [pH]	Temp. [°C]	dE [pH]
1	4.686	23.7	
2	4.936	23.8	0.250
3	5.171	24.0	0.235
4	5.357	24.3	0.186
5	5.502	24.5	0.145
6	5.625	24.7	0.123
7	5.729	24.8	0.104
8	5.813	24.9	0.084
9	5.898	25.0	0.085
10	5.977	25.0	0.079
11	6.049	25.1	0.072
12	6.117	25.1	0.068
13	6.179	25.2	0.062
14	6.241	25.2	0.062
15	6.293	25.2	0.052
16	6.348	25.2	0.055
17	6.400	25.2	0.052
18	6.450	25.2	0.050
19	6.498	25.2	0.048
20	6.543	25.3	0.045
21	6.587	25.3	0.044
22	6.625	25.3	0.038
23	6.665	25.3	0.040
24	6.705	25.3	0.040
25	6.742	25.3	0.037
26	6.778	25.3	0.036
27	6.814	25.3	0.036
28	6.849	25.3	0.035
29	6.880	25.3	0.031
30	6.908	25.3	0.028
31	6.936	25.3	0.028
32	6.964	25.3	0.028
33	6.991	25.3	0.027
34	7.014	25.3	0.023
35	7.034	25.3	0.020
36	7.061	25.3	0.027

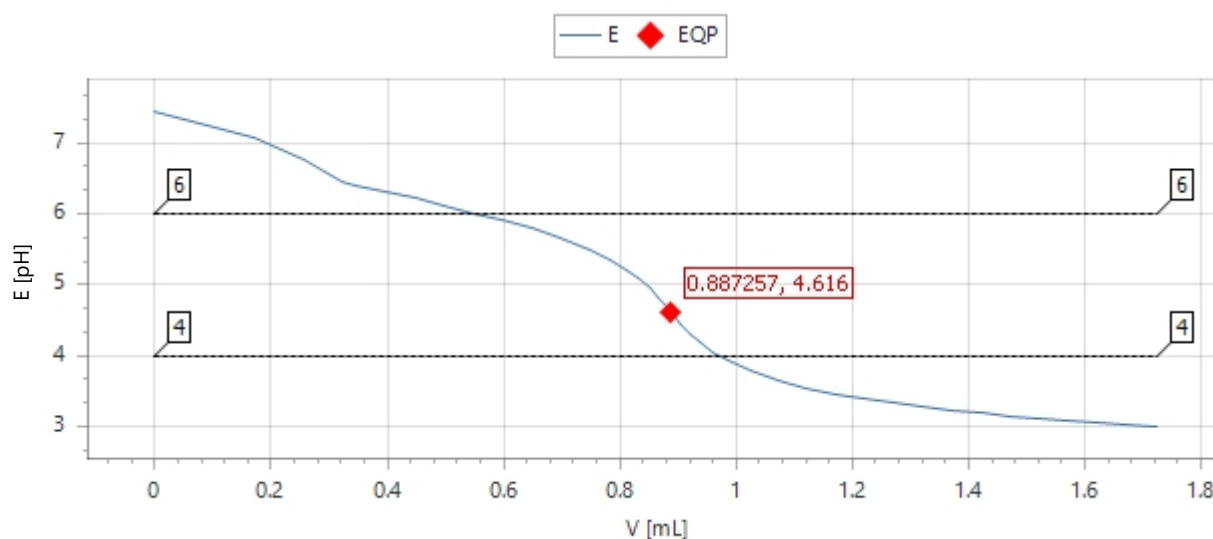
Titration report

LabX

37	7.085	25.3	0.024
38	7.109	25.3	0.024
39	7.130	25.3	0.021
40	7.152	25.3	0.022
41	7.172	25.3	0.020
42	7.190	25.3	0.018
43	7.206	25.3	0.016
44	7.224	25.3	0.018
45	7.241	25.3	0.017
46	7.257	25.3	0.016
47	7.272	25.3	0.015
48	7.287	25.3	0.015
49	7.302	25.3	0.015
50	7.314	25.3	0.012
51	7.327	25.3	0.013
52	7.340	25.3	0.013
53	7.353	25.3	0.013
54	7.364	25.3	0.011
55	7.375	25.3	0.011
56	7.386	25.3	0.011
57	7.396	25.3	0.010
58	7.406	25.3	0.010
59	7.415	25.3	0.009
60	7.424	25.3	0.009

Titration (EQP) [1] Scope 1/1, Sample 7/17

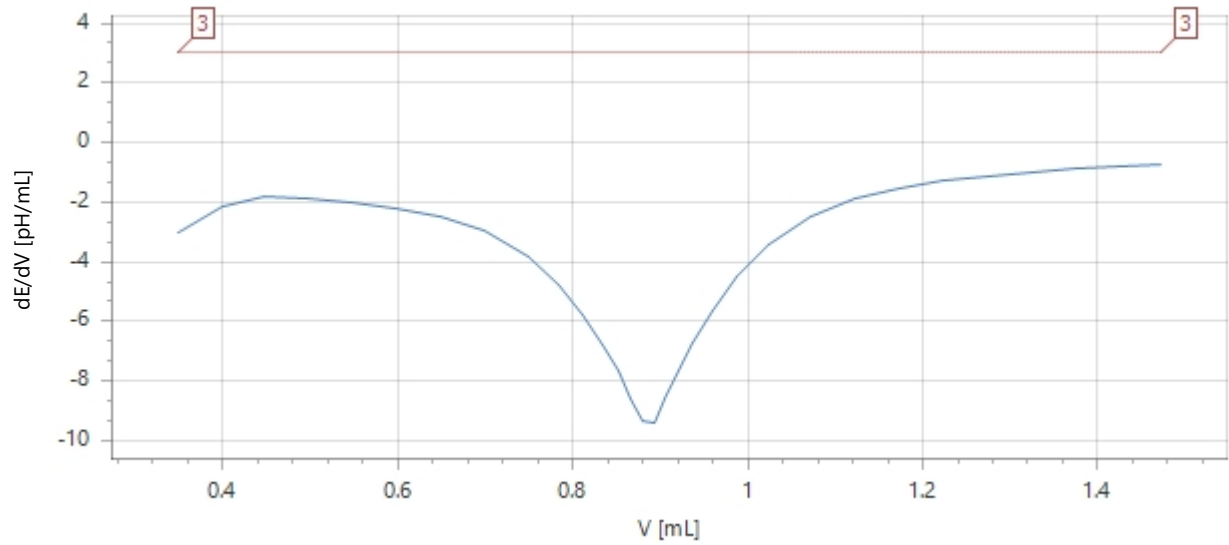
E - V Curve Scope 1/1, Sample 7/17



dE/dV - V Curve Scope 1/1, Sample 7/17

Titration report

LabX



Measured Values Scope 1/1, Sample 7/17

Time [s]	V [mL]	Meas. value [pH]	dV [mL]	Temp. [°C]	dE [pH]	dE/dV [pH/mL]	EQP
0	0.00000	7.447		25.3			
27	0.17150	7.075	0.17150	25.3	-0.372		
58	0.25725	6.754	0.08575	25.2	-0.321		
88	0.30000	6.554	0.04275	25.2	-0.200		
179	0.32325	6.437	0.02325	25.2	-0.117		
209	0.34825	6.394	0.02500	25.2	-0.043	-3.01	
239	0.39825	6.302	0.05000	25.2	-0.092	-2.15	
270	0.44825	6.213	0.05000	25.2	-0.089	-1.79	
300	0.49825	6.112	0.05000	25.2	-0.101	-1.87	
330	0.54825	6.009	0.05000	25.2	-0.103	-2.04	
360	0.59825	5.901	0.05000	25.2	-0.108	-2.22	
390	0.64825	5.789	0.05000	25.2	-0.112	-2.49	
421	0.69825	5.657	0.05000	25.2	-0.132	-2.97	
451	0.74825	5.494	0.05000	25.2	-0.163	-3.85	
481	0.78300	5.356	0.03475	25.2	-0.138	-4.78	
511	0.81075	5.220	0.02775	25.2	-0.136	-5.80	
542	0.83325	5.084	0.02250	25.2	-0.136	-6.79	
572	0.85150	4.949	0.01825	25.2	-0.135	-7.68	
602	0.86675	4.817	0.01525	25.2	-0.132	-8.60	
632	0.88000	4.689	0.01325	25.2	-0.128	-9.34	
	0.887257 ⁽⁹⁾	4.616 ⁽⁹⁾					EQP
663	0.89250	4.563	0.01250	25.2	-0.126	-9.37	
693	0.90550	4.437	0.01300	25.2	-0.126	-8.58	
723	0.92000	4.312	0.01450	25.2	-0.125	-7.63	
753	0.93775	4.180	0.01775	25.2	-0.132	-6.70	
784	0.95875	4.055	0.02100	25.2	-0.125	-5.67	
814	0.98700	3.922	0.02825	25.2	-0.133	-4.52	
844	1.02325	3.791	0.03625	25.2	-0.131	-3.44	
874	1.07275	3.656	0.04950	25.2	-0.135	-2.48	
905	1.12275	3.552	0.05000	25.1	-0.104	-1.89	
935	1.17275	3.467	0.05000	25.1	-0.085	-1.53	

Titration report

LabX

965	1.22275	3.396	0.05000	25.1	-0.071	-1.30
995	1.27275	3.334	0.05000	25.1	-0.062	-1.13
1026	1.32275	3.280	0.05000	25.1	-0.054	-1.01
1056	1.37275	3.232	0.05000	25.1	-0.048	-0.90
1086	1.42275	3.189	0.05000	25.1	-0.043	-0.82
1116	1.47275	3.150	0.05000	25.1	-0.039	-0.75
1146	1.52275	3.114	0.05000	25.1	-0.036	
1177	1.57275	3.081	0.05000	25.1	-0.033	
1207	1.62275	3.050	0.05000	25.1	-0.031	
1237	1.67275	3.021	0.05000	25.1	-0.029	
1268	1.72275	2.994	0.05000	25.1	-0.027	

Measure (normal) [1] Scope 1/1, Sample 8/17

Measured Values Scope 1/1, Sample 8/17

Time [s]	Meas. value [pH]	Temp. [°C]	dE [pH]
1	4.715	23.4	
2	4.980	23.6	0.265
3	5.197	23.9	0.217
4	5.371	24.1	0.174
5	5.514	24.4	0.143
6	5.634	24.6	0.120
7	5.737	24.7	0.103
8	5.818	24.8	0.081
9	5.900	24.9	0.082
10	5.977	25.0	0.077
11	6.048	25.1	0.071
12	6.113	25.1	0.065
13	6.175	25.2	0.062
14	6.234	25.2	0.059
15	6.289	25.2	0.055
16	6.337	25.2	0.048
17	6.387	25.2	0.050
18	6.434	25.2	0.047
19	6.480	25.2	0.046
20	6.524	25.3	0.044
21	6.566	25.3	0.042
22	6.607	25.3	0.041
23	6.643	25.3	0.036
24	6.681	25.3	0.038
25	6.716	25.3	0.035
26	6.750	25.3	0.034
27	6.783	25.3	0.033
28	6.816	25.3	0.033
29	6.846	25.3	0.030
30	6.876	25.3	0.030
31	6.902	25.3	0.026
32	6.927	25.3	0.025
33	6.953	25.3	0.026

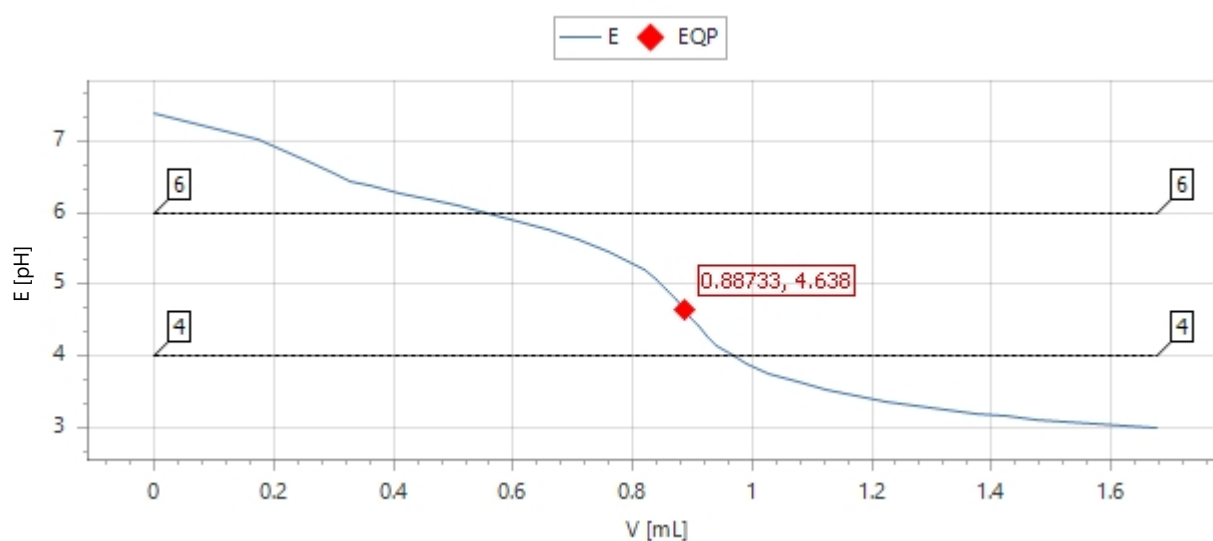
Titration report

LabX

34	6.979	25.3	0.026
35	7.003	25.3	0.024
36	7.019	25.3	0.016
37	7.039	25.3	0.020
38	7.065	25.3	0.026
39	7.086	25.3	0.021
40	7.107	25.3	0.021
41	7.127	25.3	0.020
42	7.145	25.3	0.018
43	7.161	25.3	0.016
44	7.177	25.3	0.016
45	7.193	25.3	0.016
46	7.209	25.3	0.016
47	7.223	25.3	0.014
48	7.238	25.3	0.015
49	7.252	25.3	0.014
50	7.264	25.3	0.012
51	7.277	25.3	0.013
52	7.290	25.3	0.013
53	7.302	25.3	0.012
54	7.314	25.3	0.012
55	7.325	25.3	0.011
56	7.336	25.3	0.011
57	7.345	25.3	0.009
58	7.354	25.3	0.009
59	7.363	25.3	0.009
60	7.372	25.3	0.009

Titration (EQP) [1] Scope 1/1, Sample 8/17

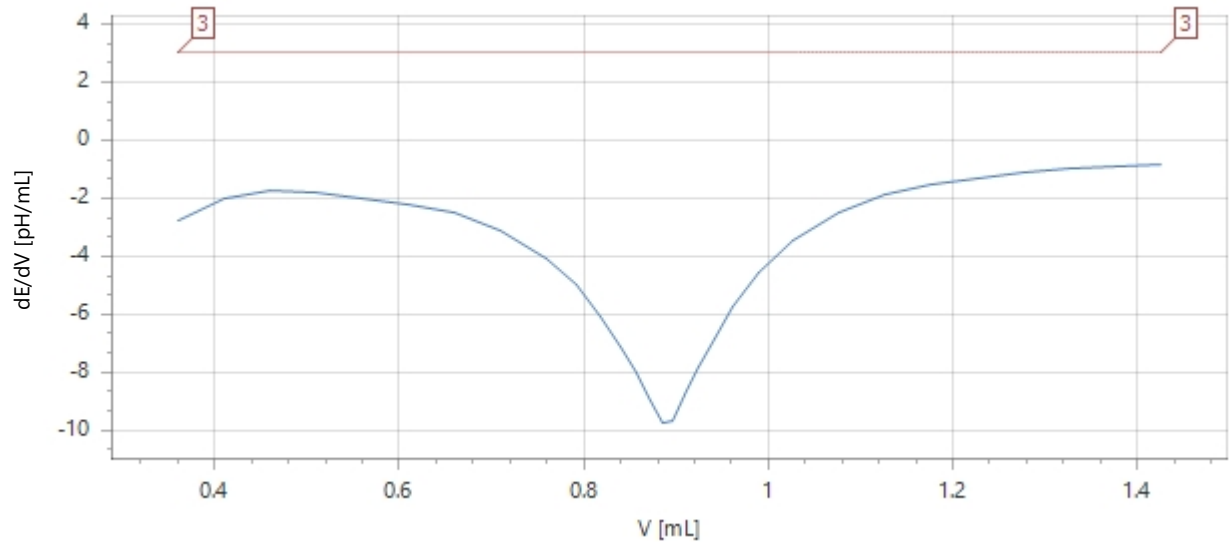
E - V Curve Scope 1/1, Sample 8/17



dE/dV - V Curve Scope 1/1, Sample 8/17

Titration report

LabX



Measured Values Scope 1/1, Sample 8/17

Time [s]	V [mL]	Meas. value [pH]	dV [mL]	Temp. [°C]	dE [pH]	dE/dV [pH/mL]	EQP
0	0.00000	7.391		25.3			
28	0.17150	7.019	0.17150	25.3	-0.372		
58	0.25725	6.718	0.08575	25.3	-0.301		
88	0.30000	6.538	0.04275	25.3	-0.180		
179	0.32700	6.426	0.02700	25.3	-0.112		
209	0.35975	6.370	0.03275	25.3	-0.056	-2.76	
239	0.40975	6.280	0.05000	25.2	-0.090	-2.03	
270	0.45975	6.179	0.05000	25.2	-0.101	-1.75	
300	0.50975	6.091	0.05000	25.2	-0.088	-1.83	
330	0.55975	5.991	0.05000	25.2	-0.100	-2.01	
360	0.60975	5.885	0.05000	25.2	-0.106	-2.20	
390	0.65975	5.767	0.05000	25.2	-0.118	-2.52	
421	0.70975	5.633	0.05000	25.2	-0.134	-3.10	
451	0.75975	5.465	0.05000	25.2	-0.168	-4.05	
481	0.79225	5.330	0.03250	25.2	-0.135	-5.00	
511	0.81875	5.195	0.02650	25.2	-0.135	-6.06	
542	0.84050	5.056	0.02175	25.2	-0.139	-7.11	
572	0.85725	4.928	0.01675	25.2	-0.128	-7.98	
602	0.87200	4.793	0.01475	25.2	-0.135	-8.97	
632	0.88450	4.668	0.01250	25.2	-0.125	-9.71	
	0.887330 ⁽⁹⁾	4.638 ⁽⁹⁾					EQP
663	0.89700	4.536	0.01250	25.2	-0.132	-9.65	
693	0.90925	4.416	0.01225	25.2	-0.120	-8.79	
723	0.92400	4.285	0.01475	25.2	-0.131	-7.79	
753	0.94100	4.157	0.01700	25.2	-0.128	-6.84	
784	0.96225	4.028	0.02125	25.2	-0.129	-5.75	
814	0.98975	3.898	0.02750	25.2	-0.130	-4.58	
844	1.02700	3.764	0.03725	25.2	-0.134	-3.45	
874	1.07625	3.630	0.04925	25.2	-0.134	-2.48	
905	1.12625	3.525	0.05000	25.2	-0.105	-1.89	
935	1.17625	3.440	0.05000	25.2	-0.085	-1.53	

Titration report

LabX

965	1.22625	3.369	0.05000	25.2	-0.071	-1.29
995	1.27625	3.308	0.05000	25.2	-0.061	-1.13
1026	1.32625	3.254	0.05000	25.2	-0.054	-1.00
1056	1.37625	3.206	0.05000	25.2	-0.048	-0.90
1086	1.42625	3.163	0.05000	25.2	-0.043	-0.82
1116	1.47625	3.124	0.05000	25.2	-0.039	
1146	1.52625	3.088	0.05000	25.2	-0.036	
1177	1.57625	3.054	0.05000	25.2	-0.034	
1207	1.62625	3.024	0.05000	25.2	-0.030	
1237	1.67625	2.995	0.05000	25.2	-0.029	

Measure (normal) [1] Scope 1/1, Sample 9/17

Measured Values Scope 1/1, Sample 9/17

Time [s]	Meas. value [pH]	Temp. [°C]	dE [pH]
1	4.755	23.5	
2	5.009	23.7	0.254
3	5.230	24.0	0.221
4	5.399	24.2	0.169
5	5.537	24.4	0.138
6	5.653	24.6	0.116
7	5.755	24.8	0.102
8	5.837	24.9	0.082
9	5.921	25.0	0.084
10	5.998	25.0	0.077
11	6.069	25.1	0.071
12	6.136	25.1	0.067
13	6.199	25.2	0.063
14	6.258	25.2	0.059
15	6.310	25.2	0.052
16	6.364	25.2	0.054
17	6.416	25.2	0.052
18	6.464	25.2	0.048
19	6.511	25.2	0.047
20	6.556	25.3	0.045
21	6.599	25.3	0.043
22	6.641	25.3	0.042
23	6.677	25.3	0.036
24	6.714	25.3	0.037
25	6.750	25.3	0.036
26	6.786	25.3	0.036
27	6.820	25.3	0.034
28	6.855	25.3	0.035
29	6.886	25.3	0.031
30	6.911	25.3	0.025
31	6.939	25.3	0.028
32	6.967	25.3	0.028
33	6.993	25.3	0.026
34	7.016	25.3	0.023

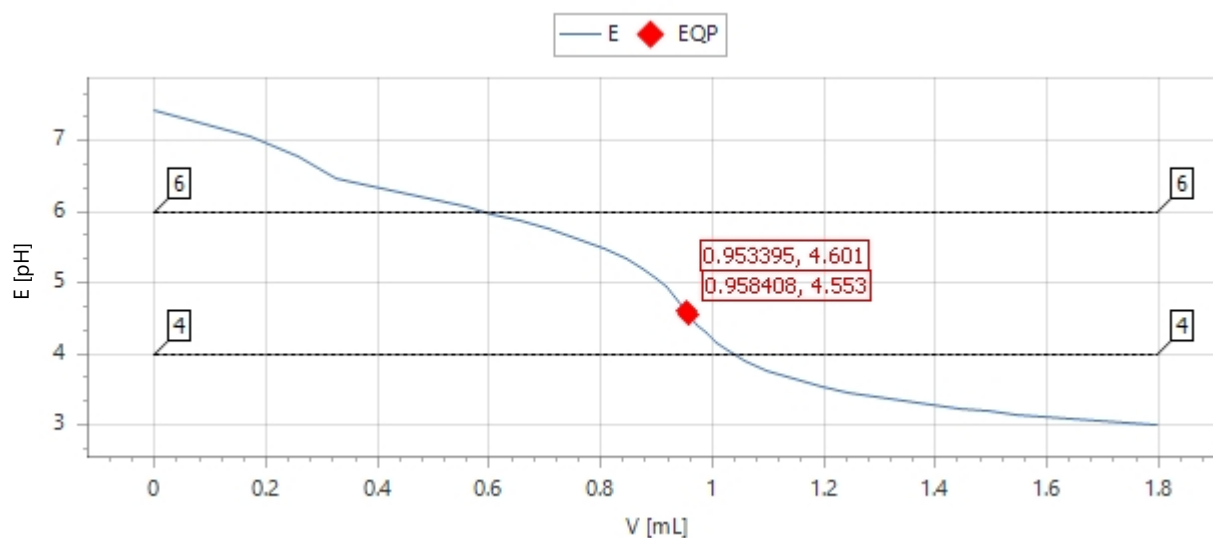
Titration report

LabX

35	7.035	25.3	0.019
36	7.061	25.3	0.026
37	7.086	25.3	0.025
38	7.109	25.3	0.023
39	7.131	25.3	0.022
40	7.151	25.3	0.020
41	7.169	25.3	0.018
42	7.188	25.3	0.019
43	7.205	25.3	0.017
44	7.221	25.3	0.016
45	7.237	25.3	0.016
46	7.253	25.3	0.016
47	7.268	25.3	0.015
48	7.283	25.3	0.015
49	7.297	25.3	0.014
50	7.309	25.3	0.012
51	7.322	25.3	0.013
52	7.335	25.3	0.013
53	7.347	25.3	0.012
54	7.358	25.3	0.011
55	7.369	25.3	0.011
56	7.379	25.3	0.010
57	7.388	25.3	0.009
58	7.397	25.3	0.009
59	7.406	25.3	0.009
60	7.416	25.3	0.010

Titration (EQP) [1] Scope 1/1, Sample 9/17

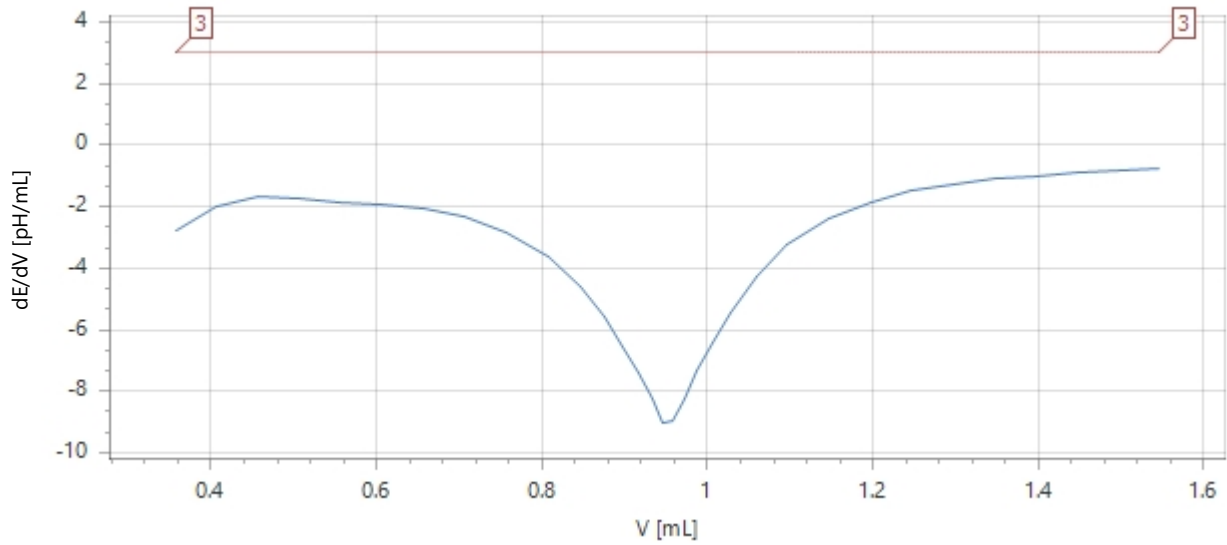
E - V Curve Scope 1/1, Sample 9/17



dE/dV - V Curve Scope 1/1, Sample 9/17

Titration report

LabX



Measured Values Scope 1/1, Sample 9/17

Time [s]	V [mL]	Meas. value [pH]	dV [mL]	Temp. [°C]	dE [pH]	dE/dV [pH/mL]	EQP
0	0.00000	7.435		25.3			
28	0.17150	7.075	0.17150	25.3	-0.360		
58	0.25725	6.773	0.08575	25.3	-0.302		
88	0.30000	6.593	0.04275	25.2	-0.180		
179	0.32725	6.474	0.02725	25.2	-0.119		
209	0.35725	6.424	0.03000	25.2	-0.050	-2.75	
239	0.40725	6.342	0.05000	25.2	-0.082	-1.99	
269	0.45725	6.253	0.05000	25.2	-0.089	-1.67	
300	0.50725	6.163	0.05000	25.2	-0.090	-1.74	
330	0.55725	6.061	0.05000	25.2	-0.102	-1.86	
360	0.60725	5.973	0.05000	25.2	-0.088	-1.94	
390	0.65725	5.873	0.05000	25.2	-0.100	-2.08	
421	0.70725	5.762	0.05000	25.2	-0.111	-2.34	
451	0.75725	5.637	0.05000	25.2	-0.125	-2.81	
481	0.80725	5.484	0.05000	25.2	-0.153	-3.64	
511	0.84650	5.335	0.03925	25.2	-0.149	-4.61	
542	0.87525	5.199	0.02875	25.2	-0.136	-5.60	
572	0.89825	5.062	0.02300	25.2	-0.137	-6.57	
602	0.91625	4.935	0.01800	25.2	-0.127	-7.38	
632	0.93250	4.800	0.01625	25.2	-0.135	-8.26	
663	0.94625	4.670	0.01375	25.2	-0.130	-9.00	
	0.953395	4.601					EQP1
	0.958408	4.553					EQP2
693	0.95875	4.550	0.01250	25.2	-0.120	-8.99	
723	0.97275	4.421	0.01400	25.2	-0.129	-8.23	
753	0.98800	4.293	0.01525	25.2	-0.128	-7.31	
784	1.00600	4.165	0.01800	25.2	-0.128	-6.43	
814	1.02825	4.039	0.02225	25.2	-0.126	-5.42	
844	1.05825	3.904	0.03000	25.2	-0.135	-4.29	
874	1.09625	3.773	0.03800	25.2	-0.131	-3.25	
904	1.14625	3.643	0.05000	25.2	-0.130	-2.36	

Titration report

LabX

935	1.19625	3.542	0.05000	25.2	-0.101	-1.83
965	1.24625	3.460	0.05000	25.2	-0.082	-1.49
995	1.29625	3.391	0.05000	25.2	-0.069	-1.27
1025	1.34625	3.330	0.05000	25.2	-0.061	-1.11
1056	1.39625	3.277	0.05000	25.2	-0.053	-0.99
1086	1.44625	3.230	0.05000	25.2	-0.047	-0.89
1116	1.49625	3.188	0.05000	25.2	-0.042	-0.81
1146	1.54625	3.149	0.05000	25.2	-0.039	-0.74
1177	1.59625	3.114	0.05000	25.2	-0.035	
1207	1.64625	3.081	0.05000	25.2	-0.033	
1237	1.69625	3.050	0.05000	25.2	-0.031	
1267	1.74625	3.022	0.05000	25.2	-0.028	
1298	1.79625	2.995	0.05000	25.2	-0.027	

Measure (normal) [1] Scope 1/1, Sample 10/17

Measured Values Scope 1/1, Sample 10/17

Time [s]	Meas. value [pH]	Temp. [°C]	dE [pH]
1	4.723	23.2	
2	5.000	23.4	0.277
3	5.243	23.7	0.243
4	5.433	24.0	0.190
5	5.587	24.3	0.154
6	5.719	24.5	0.132
7	5.837	24.7	0.118
8	5.934	24.8	0.097
9	6.033	24.9	0.099
10	6.125	25.0	0.092
11	6.213	25.1	0.088
12	6.295	25.1	0.082
13	6.373	25.2	0.078
14	6.448	25.2	0.075
15	6.512	25.2	0.064
16	6.578	25.2	0.066
17	6.641	25.2	0.063
18	6.702	25.3	0.061
19	6.760	25.3	0.058
20	6.814	25.3	0.054
21	6.867	25.3	0.053
22	6.909	25.3	0.042
23	6.954	25.3	0.045
24	6.997	25.3	0.043
25	7.033	25.3	0.036
26	7.078	25.3	0.045
27	7.117	25.3	0.039
28	7.153	25.3	0.036
29	7.185	25.3	0.032
30	7.216	25.3	0.031
31	7.246	25.3	0.030

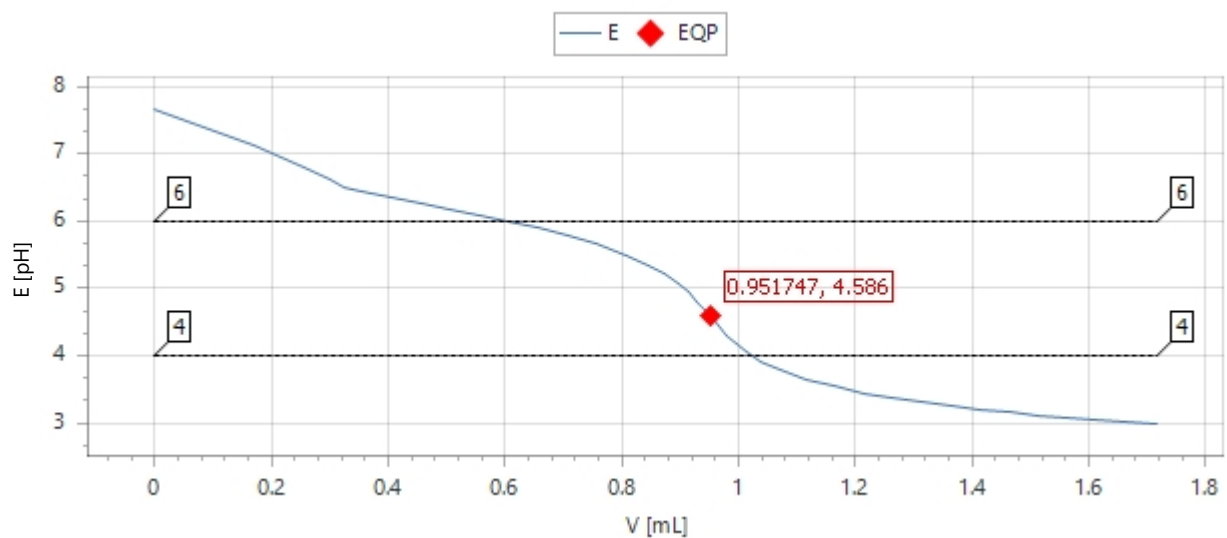
Titration report

LabX

32	7.274	25.3	0.028
33	7.300	25.3	0.026
34	7.326	25.3	0.026
35	7.350	25.3	0.024
36	7.370	25.3	0.020
37	7.391	25.3	0.021
38	7.412	25.3	0.021
39	7.431	25.3	0.019
40	7.450	25.3	0.019
41	7.467	25.3	0.017
42	7.483	25.3	0.016
43	7.496	25.3	0.013
44	7.510	25.3	0.014
45	7.524	25.3	0.014
46	7.537	25.3	0.013
47	7.549	25.3	0.012
48	7.561	25.3	0.012
49	7.572	25.3	0.011
50	7.580	25.3	0.008
51	7.590	25.3	0.010
52	7.599	25.3	0.009
53	7.608	25.3	0.009
54	7.616	25.3	0.008
55	7.624	25.3	0.008
56	7.631	25.3	0.007
57	7.638	25.3	0.007
58	7.644	25.3	0.006
59	7.651	25.3	0.007
60	7.657	25.3	0.006

Titration (EQP) [1] Scope 1/1, Sample 10/17

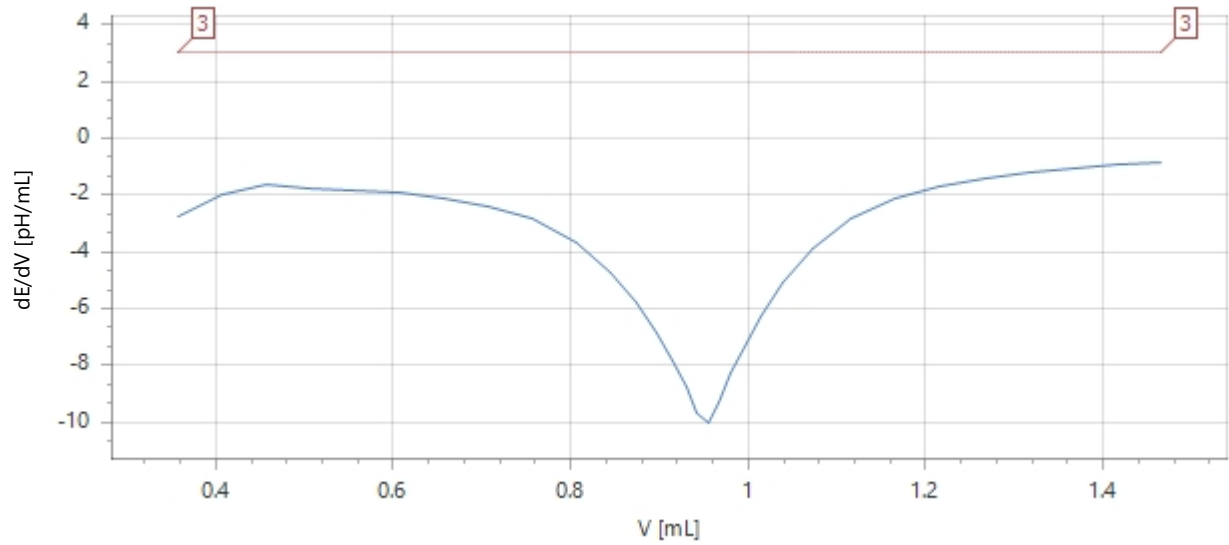
E - V Curve Scope 1/1, Sample 10/17



dE/dV - V Curve Scope 1/1, Sample 10/17

Titration report

LabX



Measured Values Scope 1/1, Sample 10/17

Time [s]	V [mL]	Meas. value [pH]	dV [mL]	Temp. [°C]	dE [pH]	dE/dV [pH/mL]	EQP
0	0.00000	7.669		25.3			
31	0.17150	7.132	0.17150	25.3	-0.537		
62	0.25725	6.804	0.08575	25.3	-0.328		
92	0.30000	6.610	0.04275	25.3	-0.194		
183	0.32500	6.500	0.02500	25.3	-0.110		
213	0.35700	6.448	0.03200	25.3	-0.052	-2.76	
243	0.40700	6.360	0.05000	25.3	-0.088	-1.98	
273	0.45700	6.271	0.05000	25.3	-0.089	-1.65	
304	0.50700	6.180	0.05000	25.3	-0.091	-1.74	
334	0.55700	6.088	0.05000	25.3	-0.092	-1.85	
364	0.60700	5.992	0.05000	25.3	-0.096	-1.94	
394	0.65700	5.892	0.05000	25.3	-0.100	-2.10	
425	0.70700	5.779	0.05000	25.3	-0.113	-2.37	
455	0.75700	5.654	0.05000	25.3	-0.125	-2.85	
485	0.80700	5.500	0.05000	25.3	-0.154	-3.69	
515	0.84575	5.350	0.03875	25.3	-0.150	-4.73	
546	0.87350	5.216	0.02775	25.3	-0.134	-5.78	
576	0.89625	5.079	0.02275	25.3	-0.137	-6.88	
606	0.91425	4.948	0.01800	25.2	-0.131	-7.83	
636	0.92950	4.812	0.01525	25.3	-0.136	-8.78	
667	0.94200	4.688	0.01250	25.3	-0.124	-9.69	
	0.951747 ⁽⁹⁾	4.586 ⁽⁹⁾					EQP
697	0.95450	4.557	0.01250	25.3	-0.131	-9.99	
727	0.96650	4.430	0.01200	25.3	-0.127	-9.30	
757	0.97950	4.308	0.01300	25.3	-0.122	-8.29	
788	0.99550	4.177	0.01600	25.3	-0.131	-7.32	
818	1.01450	4.050	0.01900	25.3	-0.127	-6.26	
848	1.03950	3.920	0.02500	25.2	-0.130	-5.07	
878	1.07250	3.786	0.03300	25.2	-0.134	-3.88	
908	1.11575	3.654	0.04325	25.2	-0.132	-2.85	
939	1.16575	3.538	0.05000	25.2	-0.116	-2.11	

Titration report

LabX

969	1.21575	3.445	0.05000	25.2	-0.093	-1.67
999	1.26575	3.369	0.05000	25.2	-0.076	-1.38
1029	1.31575	3.304	0.05000	25.2	-0.065	-1.20
1060	1.36575	3.247	0.05000	25.2	-0.057	-1.06
1090	1.41575	3.196	0.05000	25.2	-0.051	-0.95
1120	1.46575	3.151	0.05000	25.2	-0.045	-0.86
1150	1.51575	3.110	0.05000	25.2	-0.041	
1181	1.56575	3.073	0.05000	25.2	-0.037	
1211	1.61575	3.039	0.05000	25.2	-0.034	
1241	1.66575	3.007	0.05000	25.2	-0.032	
1272	1.71575	2.978	0.05000	25.2	-0.029	

Measure (normal) [1] Scope 1/1, Sample 11/17

Measured Values Scope 1/1, Sample 11/17

Time [s]	Meas. value [pH]	Temp. [°C]	dE [pH]
1	4.485	23.2	
2	4.883	23.4	0.398
3	5.179	23.7	0.296
4	5.400	24.0	0.221
5	5.577	24.3	0.177
6	5.728	24.5	0.151
7	5.863	24.7	0.135
8	5.985	24.8	0.122
9	6.086	24.9	0.101
10	6.193	25.0	0.107
11	6.292	25.1	0.099
12	6.386	25.1	0.094
13	6.475	25.2	0.089
14	6.559	25.2	0.084
15	6.638	25.2	0.079
16	6.705	25.2	0.067
17	6.775	25.3	0.070
18	6.844	25.3	0.069
19	6.906	25.3	0.062
20	6.963	25.3	0.057
21	7.014	25.3	0.051
22	7.067	25.3	0.053
23	7.113	25.3	0.046
24	7.159	25.3	0.046
25	7.201	25.3	0.042
26	7.240	25.3	0.039
27	7.279	25.3	0.039
28	7.314	25.3	0.035
29	7.348	25.3	0.034
30	7.376	25.3	0.028
31	7.405	25.3	0.029
32	7.434	25.3	0.029
33	7.460	25.3	0.026

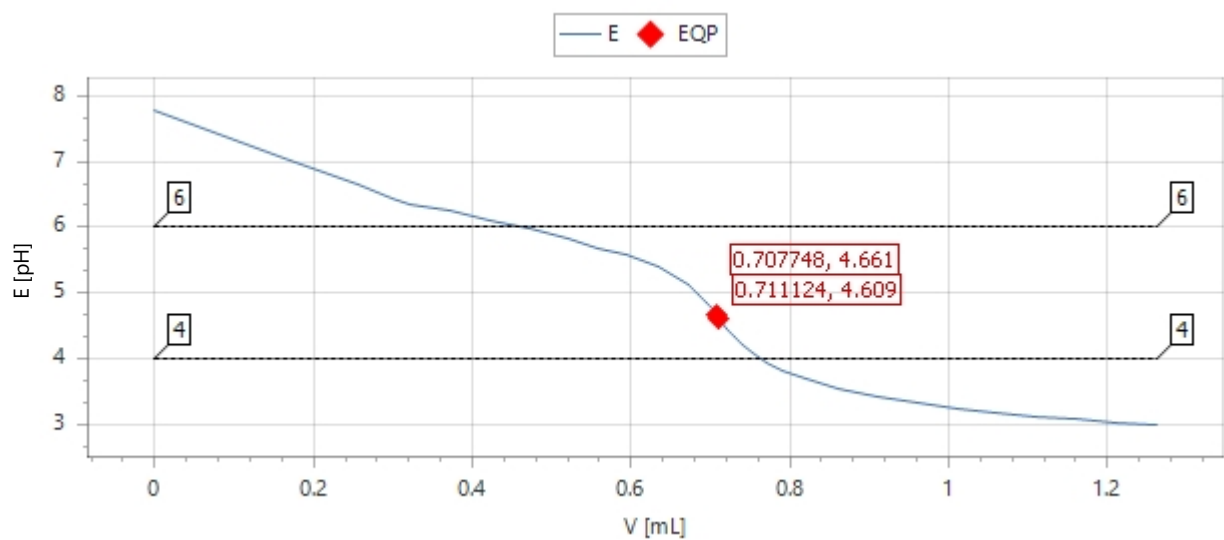
Titration report

LabX

34	7.484	25.3	0.024
35	7.507	25.3	0.023
36	7.526	25.3	0.019
37	7.546	25.3	0.020
38	7.564	25.3	0.018
39	7.582	25.3	0.018
40	7.599	25.3	0.017
41	7.614	25.3	0.015
42	7.629	25.3	0.015
43	7.641	25.3	0.012
44	7.655	25.3	0.014
45	7.666	25.3	0.011
46	7.677	25.3	0.011
47	7.688	25.3	0.011
48	7.698	25.3	0.010
49	7.708	25.3	0.010
50	7.716	25.3	0.008
51	7.724	25.3	0.008
52	7.732	25.3	0.008
53	7.740	25.3	0.008
54	7.747	25.3	0.007
55	7.753	25.3	0.006
56	7.759	25.3	0.006
57	7.764	25.3	0.005
58	7.770	25.3	0.006
59	7.775	25.3	0.005
60	7.780	25.3	0.005

Titration (EQP) [1] Scope 1/1, Sample 11/17

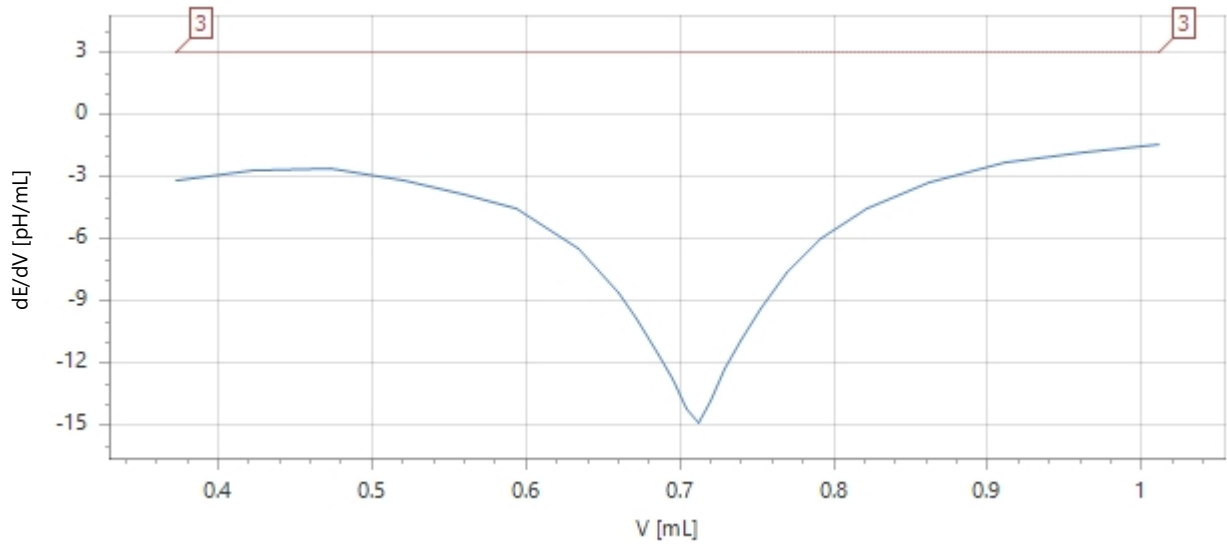
E - V Curve Scope 1/1, Sample 11/17



dE/dV - V Curve Scope 1/1, Sample 11/17

Titration report

LabX



Measured Values Scope 1/1, Sample 11/17

Time [s]	V [mL]	Meas. value [pH]	dV [mL]	Temp. [°C]	dE [pH]	dE/dV [pH/mL]	EQP
0	0.00000	7.790		25.3			
37	0.17150	7.024	0.17150	25.3	-0.766		
67	0.25725	6.654	0.08575	25.3	-0.370		
97	0.30000	6.435	0.04275	25.3	-0.219		
189	0.32250	6.354	0.02250	25.3	-0.081		
219	0.37250	6.245	0.05000	25.3	-0.109	-3.15	
249	0.42250	6.117	0.05000	25.3	-0.128	-2.63	
279	0.47250	5.979	0.05000	25.3	-0.138	-2.57	
310	0.52050	5.831	0.04800	25.3	-0.148	-3.16	
340	0.56000	5.691	0.03950	25.3	-0.140	-3.80	
370	0.59325	5.578	0.03325	25.3	-0.113	-4.51	
400	0.63475	5.401	0.04150	25.3	-0.177	-6.45	
431	0.66025	5.220	0.02550	25.3	-0.181	-8.59	
461	0.67175	5.125	0.01150	25.3	-0.095	-9.80	
491	0.68600	4.970	0.01425	25.3	-0.155	-11.58	
521	0.69550	4.846	0.00950	25.3	-0.124	-12.75	
552	0.70425	4.716	0.00875	25.3	-0.130	-14.19	
	0.707748	4.661					EQP2
	0.711124	4.609					EQP1
582	0.71225	4.591	0.00800	25.3	-0.125	-14.83	
612	0.72050	4.464	0.00825	25.3	-0.127	-13.77	
642	0.72925	4.340	0.00875	25.3	-0.124	-12.28	
673	0.73975	4.214	0.01050	25.3	-0.126	-10.88	
703	0.75300	4.084	0.01325	25.3	-0.130	-9.33	
733	0.77000	3.952	0.01700	25.3	-0.132	-7.63	
763	0.79200	3.819	0.02200	25.3	-0.133	-5.98	
794	0.82075	3.689	0.02875	25.3	-0.130	-4.52	
824	0.86150	3.553	0.04075	25.3	-0.136	-3.22	
854	0.91150	3.426	0.05000	25.3	-0.127	-2.29	
884	0.96150	3.328	0.05000	25.3	-0.098	-1.76	
914	1.01150	3.249	0.05000	25.3	-0.079	-1.44	

Titration report

LabX

945	1.06150	3.181	0.05000	25.3	-0.068
975	1.11150	3.123	0.05000	25.3	-0.058
1005	1.16150	3.071	0.05000	25.3	-0.052
1035	1.21150	3.025	0.05000	25.3	-0.046
1066	1.26150	2.984	0.05000	25.3	-0.041

Measure (normal) [1] Scope 1/1, Sample 12/17

Measured Values Scope 1/1, Sample 12/17

Time [s]	Meas. value [pH]	Temp. [°C]	dE [pH]
1	3.573	18.9	
2	4.834	19.6	1.261
3	5.197	20.5	0.363
4	5.420	21.3	0.223
5	5.595	22.1	0.175
6	5.743	22.8	0.148
7	5.873	23.3	0.130
8	5.979	23.6	0.106
9	6.088	24.0	0.109
10	6.190	24.2	0.102
11	6.285	24.4	0.095
12	6.375	24.6	0.090
13	6.459	24.7	0.084
14	6.529	24.8	0.070
15	6.602	24.8	0.073
16	6.672	24.9	0.070
17	6.736	24.9	0.064
18	6.797	25.0	0.061
19	6.858	25.0	0.061
20	6.912	25.0	0.054
21	6.957	25.0	0.045
22	7.004	25.0	0.047
23	7.047	25.0	0.043
24	7.095	25.1	0.048
25	7.137	25.1	0.042
26	7.175	25.1	0.038
27	7.207	25.1	0.032
28	7.241	25.1	0.034
29	7.273	25.1	0.032
30	7.302	25.1	0.029
31	7.330	25.1	0.028
32	7.356	25.1	0.026
33	7.379	25.1	0.023
34	7.402	25.1	0.023
35	7.423	25.1	0.021
36	7.443	25.1	0.020
37	7.462	25.1	0.019
38	7.479	25.1	0.017
39	7.496	25.1	0.017

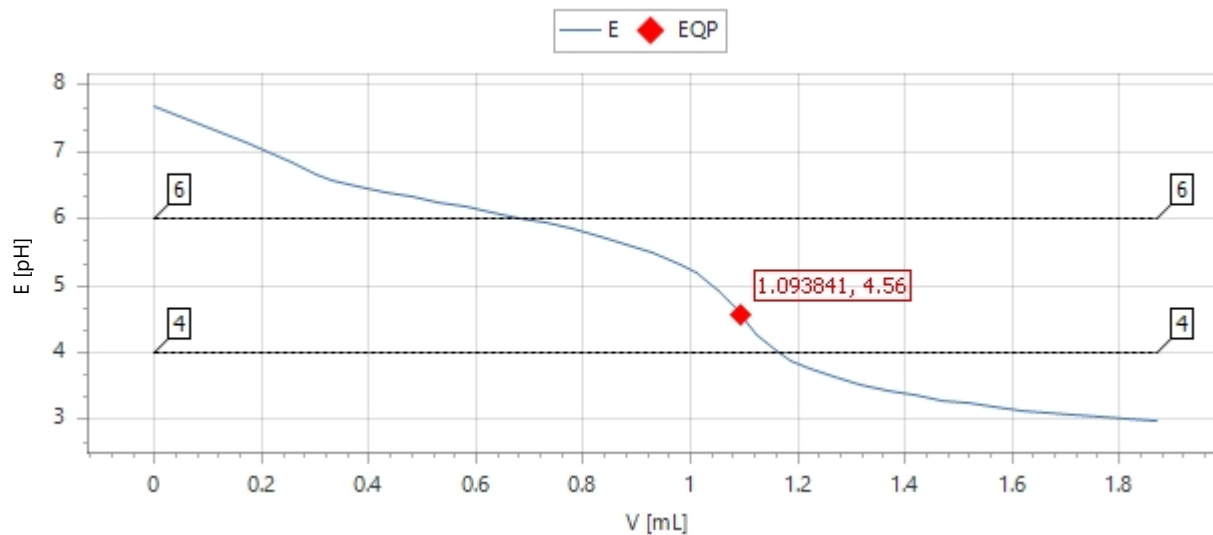
Titration report

LabX

40	7.510	25.1	0.014
41	7.525	25.1	0.015
42	7.539	25.1	0.014
43	7.552	25.1	0.013
44	7.563	25.1	0.011
45	7.575	25.1	0.012
46	7.586	25.1	0.011
47	7.595	25.1	0.009
48	7.604	25.1	0.009
49	7.615	25.1	0.011
50	7.623	25.1	0.008
51	7.631	25.1	0.008
52	7.637	25.1	0.006
53	7.644	25.1	0.007
54	7.650	25.1	0.006
55	7.657	25.1	0.007
56	7.662	25.1	0.005
57	7.668	25.1	0.006
58	7.673	25.1	0.005
59	7.677	25.1	0.004
60	7.682	25.1	0.005

Titration (EQP) [1] Scope 1/1, Sample 12/17

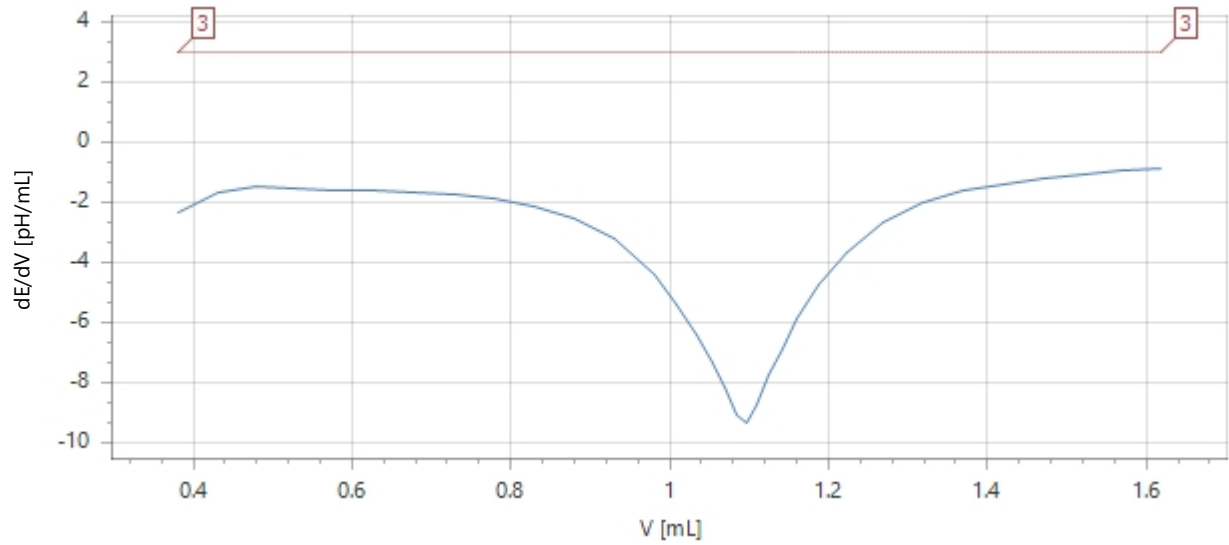
E - V Curve Scope 1/1, Sample 12/17



dE/dV - V Curve Scope 1/1, Sample 12/17

Titration report

LabX



Measured Values Scope 1/1, Sample 12/17

Time [s]	V [mL]	Meas. value [pH]	dV [mL]	Temp. [°C]	dE [pH]	dE/dV [pH/mL]	EQP
0	0.00000	7.692		25.1			
29	0.17150	7.154	0.17150	25.1	-0.538		
59	0.25725	6.838	0.08575	25.1	-0.316		
89	0.30000	6.665	0.04275	25.1	-0.173		
181	0.33050	6.568	0.03050	25.1	-0.097		
211	0.38050	6.491	0.05000	25.1	-0.077	-2.31	
241	0.43050	6.407	0.05000	25.1	-0.084	-1.65	
271	0.48050	6.330	0.05000	25.1	-0.077	-1.44	
302	0.53050	6.249	0.05000	25.1	-0.081	-1.51	
332	0.58050	6.173	0.05000	25.1	-0.076	-1.55	
362	0.63050	6.095	0.05000	25.1	-0.078	-1.56	
392	0.68050	6.015	0.05000	25.1	-0.080	-1.61	
423	0.73050	5.933	0.05000	25.1	-0.082	-1.68	
453	0.78050	5.843	0.05000	25.1	-0.090	-1.85	
483	0.83050	5.745	0.05000	25.1	-0.098	-2.10	
513	0.88050	5.632	0.05000	25.1	-0.113	-2.51	
544	0.93050	5.496	0.05000	25.2	-0.136	-3.20	
574	0.98000	5.321	0.04950	25.2	-0.175	-4.35	
604	1.00975	5.191	0.02975	25.2	-0.130	-5.35	
634	1.03450	5.053	0.02475	25.2	-0.138	-6.40	
665	1.05375	4.920	0.01925	25.2	-0.133	-7.28	
695	1.06950	4.794	0.01575	25.2	-0.126	-8.08	
725	1.08400	4.657	0.01450	25.2	-0.137	-9.03	
	1.093841 ⁽⁹⁾	4.560 ⁽⁹⁾					EQP
755	1.09625	4.536	0.01225	25.2	-0.121	-9.28	
786	1.10950	4.408	0.01325	25.2	-0.128	-8.68	
816	1.12375	4.284	0.01425	25.2	-0.124	-7.74	
846	1.14100	4.152	0.01725	25.2	-0.132	-6.82	
876	1.16100	4.027	0.02000	25.2	-0.125	-5.85	
906	1.18775	3.894	0.02675	25.2	-0.133	-4.72	
937	1.22225	3.764	0.03450	25.2	-0.130	-3.63	

Titration report

LabX

967	1.26900	3.630	0.04675	25.2	-0.134	-2.64
997	1.31900	3.520	0.05000	25.2	-0.110	-2.00
1028	1.36900	3.431	0.05000	25.2	-0.089	-1.60
1058	1.41900	3.357	0.05000	25.2	-0.074	-1.34
1088	1.46900	3.294	0.05000	25.2	-0.063	-1.16
1118	1.51900	3.239	0.05000	25.2	-0.055	-1.03
1148	1.56900	3.190	0.05000	25.2	-0.049	-0.92
1179	1.61900	3.146	0.05000	25.2	-0.044	-0.83
1209	1.66900	3.106	0.05000	25.2	-0.040	
1239	1.71900	3.070	0.05000	25.2	-0.036	
1269	1.76900	3.036	0.05000	25.2	-0.034	
1300	1.81900	3.004	0.05000	25.2	-0.032	
1330	1.86900	2.975	0.05000	25.2	-0.029	

Measure (normal) [1] Scope 1/1, Sample 13/17

Measured Values Scope 1/1, Sample 13/17

Time [s]	Meas. value [pH]	Temp. [°C]	dE [pH]
1	4.686	23.4	
2	4.966	23.6	0.280
3	5.227	23.9	0.261
4	5.435	24.1	0.208
5	5.605	24.4	0.170
6	5.749	24.6	0.144
7	5.877	24.8	0.128
8	5.994	24.9	0.117
9	6.092	25.0	0.098
10	6.194	25.1	0.102
11	6.294	25.1	0.100
12	6.387	25.2	0.093
13	6.473	25.2	0.086
14	6.554	25.2	0.081
15	6.624	25.2	0.070
16	6.696	25.3	0.072
17	6.764	25.3	0.068
18	6.828	25.3	0.064
19	6.887	25.3	0.059
20	6.938	25.3	0.051
21	6.990	25.3	0.052
22	7.030	25.3	0.040
23	7.082	25.3	0.052
24	7.128	25.3	0.046
25	7.170	25.3	0.042
26	7.207	25.3	0.037
27	7.242	25.3	0.035
28	7.276	25.3	0.034
29	7.308	25.3	0.032
30	7.334	25.3	0.026
31	7.362	25.3	0.028

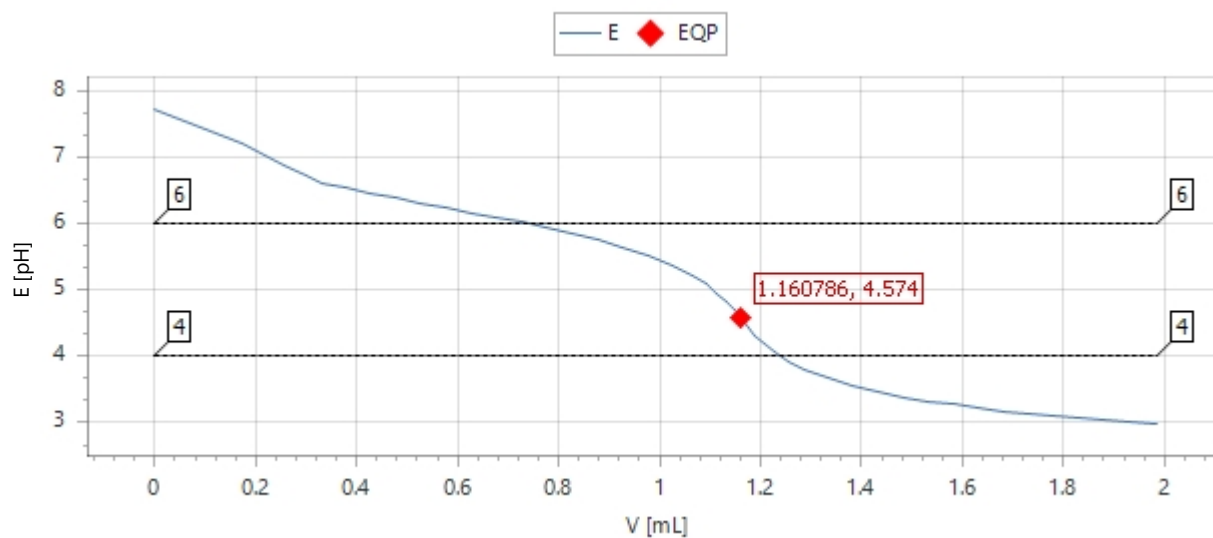
Titration report

LabX

32	7.388	25.3	0.026
33	7.412	25.3	0.024
34	7.435	25.3	0.023
35	7.457	25.3	0.022
36	7.476	25.3	0.019
37	7.495	25.3	0.019
38	7.513	25.3	0.018
39	7.528	25.3	0.015
40	7.546	25.3	0.018
41	7.560	25.3	0.014
42	7.573	25.3	0.013
43	7.585	25.3	0.012
44	7.597	25.3	0.012
45	7.609	25.3	0.012
46	7.621	25.3	0.012
47	7.631	25.3	0.010
48	7.640	25.3	0.009
49	7.649	25.3	0.009
50	7.656	25.3	0.007
51	7.664	25.3	0.008
52	7.671	25.3	0.007
53	7.678	25.3	0.007
54	7.684	25.3	0.006
55	7.690	25.3	0.006
56	7.696	25.3	0.006
57	7.701	25.3	0.005
58	7.706	25.3	0.005
59	7.710	25.3	0.004
60	7.714	25.3	0.004

Titration (EQP) [1] Scope 1/1, Sample 13/17

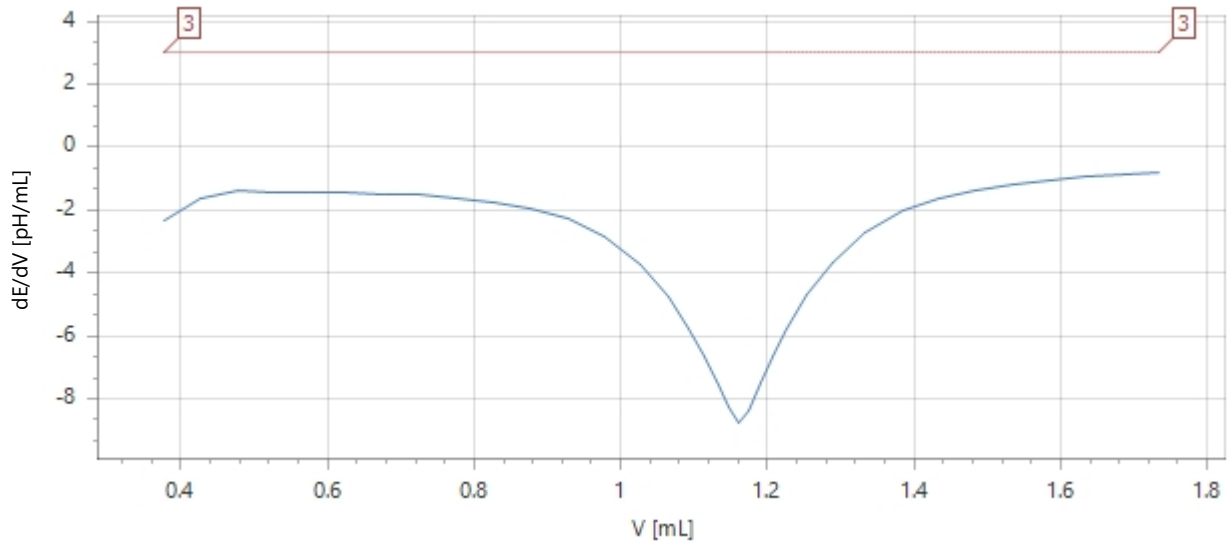
E - V Curve Scope 1/1, Sample 13/17



dE/dV - V Curve Scope 1/1, Sample 13/17

Titration report

LabX



Measured Values Scope 1/1, Sample 13/17

Time [s]	V [mL]	Meas. value [pH]	dV [mL]	Temp. [°C]	dE [pH]	dE/dV [pH/mL]	EQP
0	0.00000	7.723		25.3			
28	0.17150	7.198	0.17150	25.3	-0.525		
58	0.25725	6.884	0.08575	25.3	-0.314		
88	0.30000	6.712	0.04275	25.3	-0.172		
179	0.33100	6.607	0.03100	25.2	-0.105		
209	0.37750	6.540	0.04650	25.2	-0.067	-2.32	
239	0.42750	6.460	0.05000	25.2	-0.080	-1.63	
269	0.47750	6.380	0.05000	25.2	-0.080	-1.38	
300	0.52750	6.307	0.05000	25.2	-0.073	-1.42	
330	0.57750	6.233	0.05000	25.2	-0.074	-1.46	
360	0.62750	6.162	0.05000	25.2	-0.071	-1.45	
390	0.67750	6.088	0.05000	25.2	-0.074	-1.47	
421	0.72750	6.013	0.05000	25.2	-0.075	-1.53	
451	0.77750	5.933	0.05000	25.1	-0.080	-1.61	
481	0.82750	5.849	0.05000	25.1	-0.084	-1.74	
511	0.87750	5.757	0.05000	25.1	-0.092	-1.95	
542	0.92750	5.651	0.05000	25.1	-0.106	-2.28	
572	0.97750	5.529	0.05000	25.1	-0.122	-2.83	
602	1.02750	5.373	0.05000	25.1	-0.156	-3.74	
632	1.06500	5.227	0.03750	25.1	-0.146	-4.76	
663	1.09300	5.087	0.02800	25.1	-0.140	-5.77	
693	1.11425	4.956	0.02125	25.1	-0.131	-6.64	
723	1.13200	4.827	0.01775	25.1	-0.129	-7.41	
753	1.14775	4.694	0.01575	25.0	-0.133	-8.26	
	1.160786 ⁽⁹⁾	4.574 ⁽⁹⁾					EQP
784	1.16150	4.567	0.01375	25.0	-0.127	-8.74	
814	1.17500	4.443	0.01350	25.0	-0.124	-8.36	
844	1.18975	4.317	0.01475	25.0	-0.126	-7.50	
874	1.20700	4.186	0.01725	25.0	-0.131	-6.65	
904	1.22725	4.062	0.02025	25.0	-0.124	-5.76	
935	1.25425	3.929	0.02700	25.0	-0.133	-4.69	

Titration report

LabX

965	1.28850	3.797	0.03425	25.0	-0.132	-3.64
995	1.33375	3.666	0.04525	25.0	-0.131	-2.68
1025	1.38375	3.554	0.05000	25.0	-0.112	-2.02
1056	1.43375	3.465	0.05000	25.0	-0.089	-1.61
1086	1.48375	3.391	0.05000	25.0	-0.074	-1.35
1116	1.53375	3.327	0.05000	25.0	-0.064	-1.17
1146	1.58375	3.271	0.05000	25.0	-0.056	-1.04
1177	1.63375	3.221	0.05000	25.0	-0.050	-0.93
1207	1.68375	3.177	0.05000	25.0	-0.044	-0.84
1237	1.73375	3.137	0.05000	25.0	-0.040	-0.77
1267	1.78375	3.100	0.05000	25.0	-0.037	
1298	1.83375	3.066	0.05000	25.0	-0.034	
1328	1.88375	3.034	0.05000	25.0	-0.032	
1358	1.93375	3.005	0.05000	25.0	-0.029	
1388	1.98375	2.977	0.05000	25.0	-0.028	

Measure (normal) [1] Scope 1/1, Sample 14/17

Measured Values Scope 1/1, Sample 14/17

Time [s]	Meas. value [pH]	Temp. [°C]	dE [pH]
1	4.725	23.4	
2	5.006	23.6	0.281
3	5.272	23.9	0.266
4	5.478	24.1	0.206
5	5.647	24.4	0.169
6	5.795	24.6	0.148
7	5.928	24.7	0.133
8	6.041	24.8	0.113
9	6.156	24.9	0.115
10	6.266	25.0	0.110
11	6.370	25.1	0.104
12	6.466	25.1	0.096
13	6.554	25.1	0.088
14	6.639	25.2	0.085
15	6.718	25.2	0.079
16	6.785	25.2	0.067
17	6.857	25.2	0.072
18	6.922	25.2	0.065
19	6.984	25.2	0.062
20	7.036	25.2	0.052
21	7.094	25.2	0.058
22	7.141	25.2	0.047
23	7.188	25.2	0.047
24	7.233	25.2	0.045
25	7.274	25.2	0.041
26	7.312	25.2	0.038
27	7.348	25.3	0.036
28	7.380	25.3	0.032
29	7.409	25.3	0.029

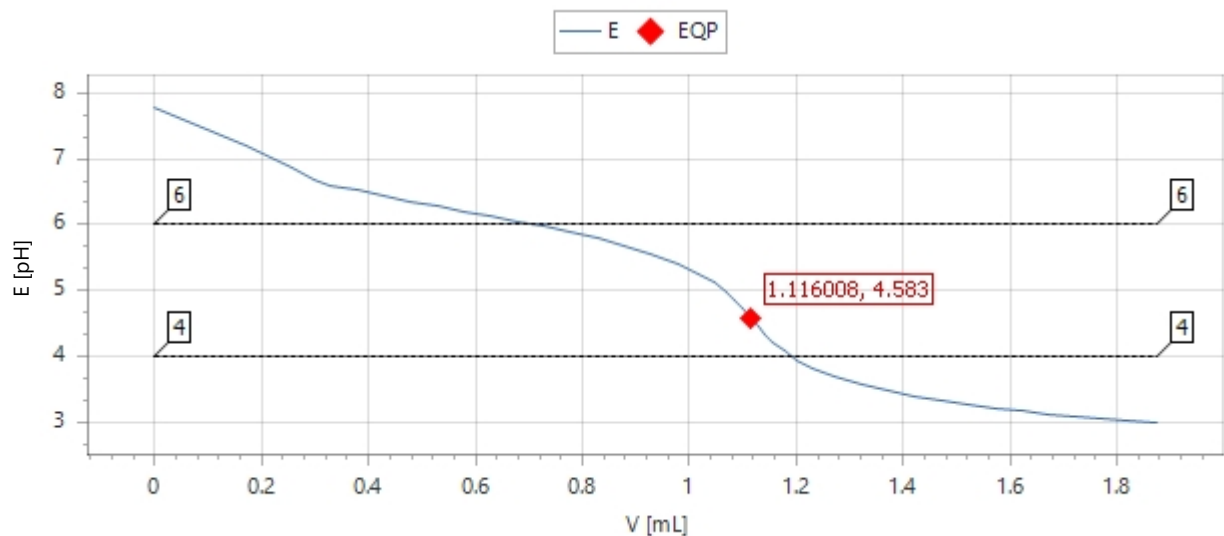
Titration report

LabX

30	7.438	25.3	0.029
31	7.466	25.3	0.028
32	7.491	25.3	0.025
33	7.515	25.3	0.024
34	7.536	25.3	0.021
35	7.555	25.3	0.019
36	7.572	25.3	0.017
37	7.590	25.3	0.018
38	7.606	25.3	0.016
39	7.621	25.3	0.015
40	7.635	25.3	0.014
41	7.648	25.3	0.013
42	7.660	25.3	0.012
43	7.670	25.3	0.010
44	7.681	25.3	0.011
45	7.691	25.3	0.010
46	7.700	25.3	0.009
47	7.709	25.3	0.009
48	7.717	25.3	0.008
49	7.724	25.3	0.007
50	7.730	25.3	0.006
51	7.737	25.3	0.007
52	7.743	25.3	0.006
53	7.749	25.3	0.006
54	7.754	25.3	0.005
55	7.759	25.3	0.005
56	7.764	25.3	0.005
57	7.767	25.3	0.003
58	7.772	25.3	0.005
59	7.776	25.3	0.004
60	7.779	25.3	0.003

Titration (EQP) [1] Scope 1/1, Sample 14/17

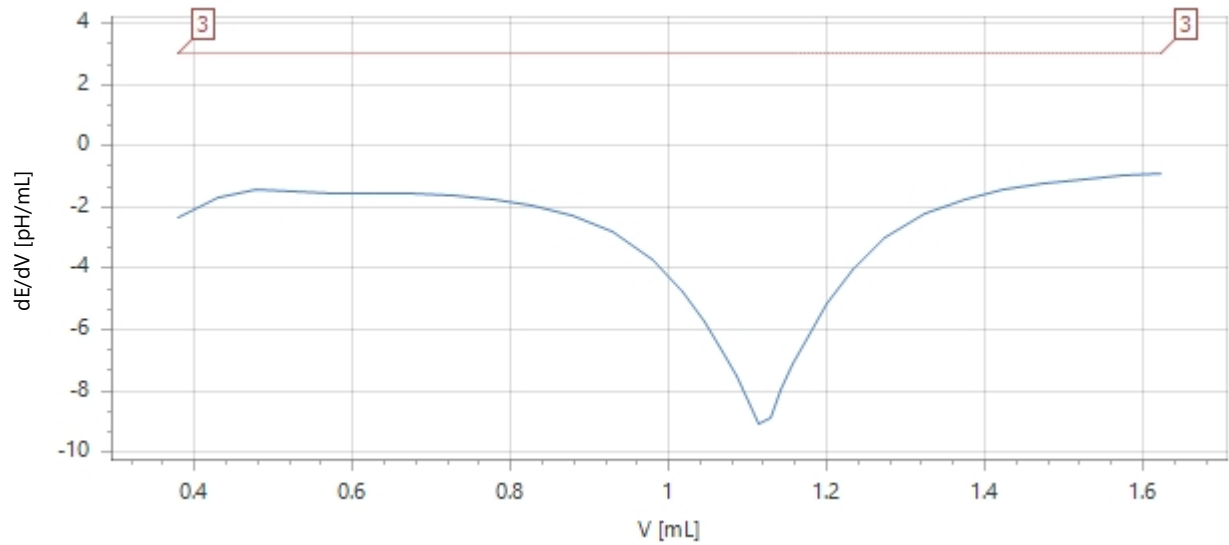
E - V Curve Scope 1/1, Sample 14/17



Titration report

LabX

dE/dV - V Curve Scope 1/1, Sample 14/17



Measured Values Scope 1/1, Sample 14/17

Time [s]	V [mL]	Meas. value [pH]	dV [mL]	Temp. [°C]	dE [pH]	dE/dV [pH/mL]	EQP
0	0.00000	7.785		25.3			
30	0.17150	7.188	0.17150	25.2	-0.597		
60	0.25725	6.872	0.08575	25.2	-0.316		
90	0.30000	6.697	0.04275	25.2	-0.175		
181	0.33000	6.597	0.03000	25.2	-0.100		
211	0.37875	6.523	0.04875	25.2	-0.074	-2.31	
241	0.42875	6.443	0.05000	25.2	-0.080	-1.67	
271	0.47875	6.362	0.05000	25.2	-0.081	-1.45	
302	0.52875	6.280	0.05000	25.2	-0.082	-1.51	
332	0.57875	6.202	0.05000	25.2	-0.078	-1.55	
362	0.62875	6.128	0.05000	25.2	-0.074	-1.54	
392	0.67875	6.051	0.05000	25.2	-0.077	-1.56	
423	0.72875	5.969	0.05000	25.2	-0.082	-1.63	
453	0.77875	5.883	0.05000	25.2	-0.086	-1.76	
483	0.82875	5.791	0.05000	25.2	-0.092	-1.96	
513	0.87875	5.684	0.05000	25.2	-0.107	-2.27	
544	0.92875	5.562	0.05000	25.2	-0.122	-2.80	
574	0.97875	5.410	0.05000	25.2	-0.152	-3.68	
604	1.01850	5.256	0.03975	25.2	-0.154	-4.76	
634	1.04600	5.121	0.02750	25.2	-0.135	-5.77	
664	1.06800	4.985	0.02200	25.2	-0.136	-6.72	
695	1.08550	4.857	0.01750	25.2	-0.128	-7.53	
725	1.10100	4.723	0.01550	25.2	-0.134	-8.39	
755	1.11450	4.597	0.01350	25.2	-0.126	-9.05	
	1.116008 ⁽⁹⁾	4.583 ⁽⁹⁾					EQP
786	1.12775	4.471	0.01325	25.2	-0.126	-8.85	
816	1.14200	4.340	0.01425	25.2	-0.131	-7.98	
846	1.15700	4.220	0.01500	25.2	-0.120	-7.13	
876	1.17650	4.087	0.01950	25.2	-0.133	-6.19	

Titration report

LabX

906	1.20000	3.959	0.02350	25.2	-0.128	-5.16
937	1.23150	3.827	0.03150	25.2	-0.132	-4.04
967	1.27325	3.692	0.04175	25.2	-0.135	-3.00
997	1.32325	3.571	0.05000	25.2	-0.121	-2.21
1027	1.37325	3.476	0.05000	25.2	-0.095	-1.74
1058	1.42325	3.396	0.05000	25.2	-0.080	-1.44
1088	1.47325	3.328	0.05000	25.2	-0.068	-1.23
1118	1.52325	3.270	0.05000	25.2	-0.058	-1.08
1148	1.57325	3.218	0.05000	25.2	-0.052	-0.97
1179	1.62325	3.172	0.05000	25.2	-0.046	-0.87
1209	1.67325	3.131	0.05000	25.2	-0.041	
1239	1.72325	3.093	0.05000	25.2	-0.038	
1269	1.77325	3.058	0.05000	25.1	-0.035	
1300	1.82325	3.026	0.05000	25.1	-0.032	
1330	1.87325	2.995	0.05000	25.1	-0.031	

Measure (normal) [1] Scope 1/1, Sample 15/17

Measured Values Scope 1/1, Sample 15/17

Time [s]	Meas. value [pH]	Temp. [°C]	dE [pH]
1	4.800	23.5	
2	5.069	23.7	0.269
3	5.313	23.9	0.244
4	5.510	24.2	0.197
5	5.678	24.4	0.168
6	5.827	24.6	0.149
7	5.962	24.8	0.135
8	6.075	24.9	0.113
9	6.191	25.0	0.116
10	6.298	25.0	0.107
11	6.400	25.1	0.102
12	6.494	25.1	0.094
13	6.582	25.2	0.088
14	6.666	25.2	0.084
15	6.747	25.2	0.081
16	6.816	25.2	0.069
17	6.887	25.2	0.071
18	6.949	25.3	0.062
19	7.009	25.3	0.060
20	7.068	25.3	0.059
21	7.123	25.3	0.055
22	7.168	25.3	0.045
23	7.213	25.3	0.045
24	7.256	25.3	0.043
25	7.296	25.3	0.040
26	7.333	25.3	0.037
27	7.369	25.3	0.036
28	7.402	25.3	0.033
29	7.429	25.3	0.027

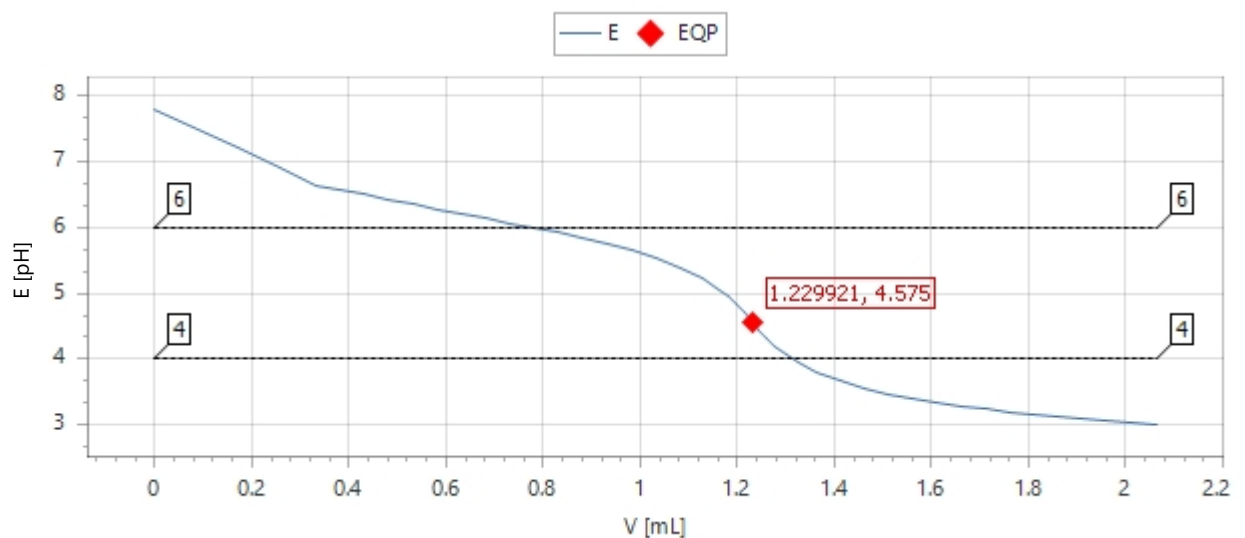
Titration report

LabX

30	7.458	25.3	0.029
31	7.484	25.3	0.026
32	7.509	25.3	0.025
33	7.533	25.3	0.024
34	7.554	25.3	0.021
35	7.574	25.3	0.020
36	7.591	25.3	0.017
37	7.609	25.3	0.018
38	7.626	25.3	0.017
39	7.641	25.3	0.015
40	7.656	25.3	0.015
41	7.669	25.3	0.013
42	7.681	25.3	0.012
43	7.692	25.3	0.011
44	7.702	25.3	0.010
45	7.711	25.3	0.009
46	7.720	25.3	0.009
47	7.729	25.3	0.009
48	7.737	25.3	0.008
49	7.743	25.3	0.006
50	7.750	25.3	0.007
51	7.756	25.3	0.006
52	7.762	25.3	0.006
53	7.767	25.3	0.005
54	7.772	25.3	0.005
55	7.776	25.3	0.004
56	7.781	25.3	0.005
57	7.784	25.3	0.003
58	7.788	25.3	0.004
59	7.792	25.3	0.004
60	7.795	25.3	0.003

Titration (EQP) [1] Scope 1/1, Sample 15/17

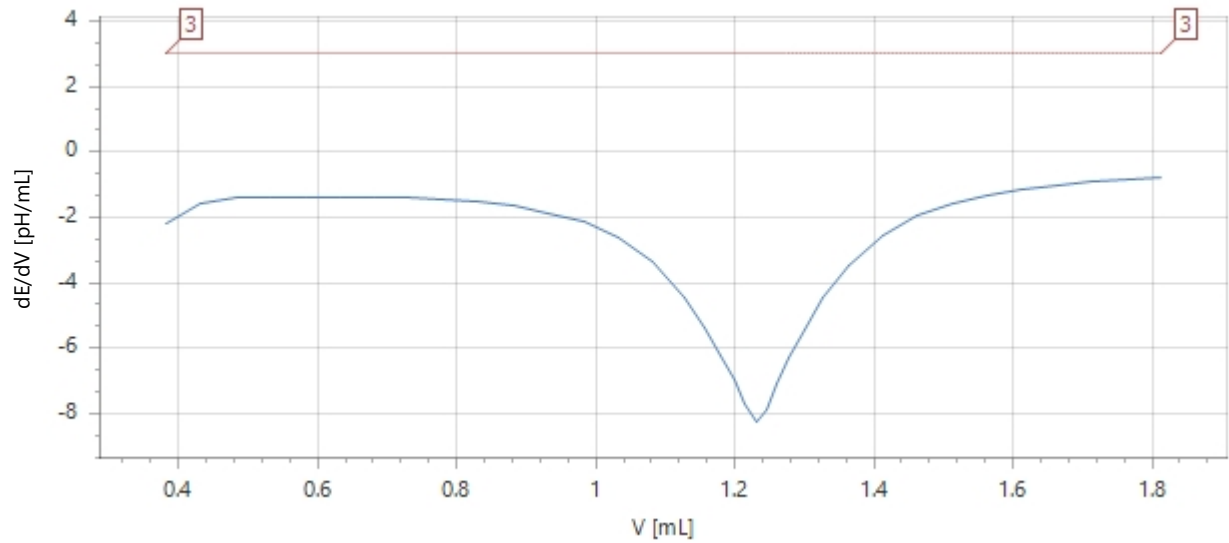
E - V Curve Scope 1/1, Sample 15/17



Titration report

LabX

dE/dV - V Curve Scope 1/1, Sample 15/17



Measured Values Scope 1/1, Sample 15/17

Time [s]	V [mL]	Meas. value [pH]	dV [mL]	Temp. [°C]	dE [pH]	dE/dV [pH/mL]	EQP
0	0.00000	7.803		25.3			
29	0.17150	7.227	0.17150	25.3	-0.576		
59	0.25725	6.918	0.08575	25.3	-0.309		
90	0.30000	6.753	0.04275	25.2	-0.165		
181	0.33250	6.649	0.03250	25.2	-0.104		
211	0.38250	6.578	0.05000	25.2	-0.071	-2.19	
241	0.43250	6.499	0.05000	25.2	-0.079	-1.57	
271	0.48250	6.423	0.05000	25.2	-0.076	-1.36	
302	0.53250	6.350	0.05000	25.2	-0.073	-1.39	
332	0.58250	6.277	0.05000	25.2	-0.073	-1.42	
362	0.63250	6.208	0.05000	25.2	-0.069	-1.39	
392	0.68250	6.140	0.05000	25.2	-0.068	-1.38	
422	0.73250	6.070	0.05000	25.2	-0.070	-1.41	
453	0.78250	5.997	0.05000	25.2	-0.073	-1.46	
483	0.83250	5.922	0.05000	25.2	-0.075	-1.54	
513	0.88250	5.842	0.05000	25.2	-0.080	-1.66	
544	0.93250	5.753	0.05000	25.2	-0.089	-1.85	
574	0.98250	5.652	0.05000	25.2	-0.101	-2.15	
604	1.03250	5.537	0.05000	25.2	-0.115	-2.62	
634	1.08250	5.395	0.05000	25.2	-0.142	-3.38	
664	1.12800	5.229	0.04550	25.2	-0.166	-4.48	
695	1.15700	5.093	0.02900	25.2	-0.136	-5.40	
725	1.17975	4.962	0.02275	25.2	-0.131	-6.23	
755	1.19850	4.832	0.01875	25.2	-0.130	-6.96	
785	1.21500	4.702	0.01650	25.2	-0.130	-7.70	
	1.229921 ⁽⁹⁾	4.575 ⁽⁹⁾					EQP
816	1.23025	4.572	0.01525	25.2	-0.130	-8.23	
846	1.24475	4.446	0.01450	25.2	-0.126	-7.90	
876	1.26025	4.321	0.01550	25.2	-0.125	-7.08	

Titration report

LabX

906	1.27850	4.191	0.01825	25.2	-0.130	-6.28
937	1.29975	4.065	0.02125	25.2	-0.126	-5.45
967	1.32750	3.935	0.02775	25.2	-0.130	-4.46
997	1.36400	3.803	0.03650	25.2	-0.132	-3.45
1027	1.41250	3.668	0.04850	25.2	-0.135	-2.53
1058	1.46250	3.561	0.05000	25.2	-0.107	-1.94
1088	1.51250	3.475	0.05000	25.2	-0.086	-1.57
1118	1.56250	3.402	0.05000	25.2	-0.073	-1.32
1148	1.61250	3.340	0.05000	25.2	-0.062	-1.14
1178	1.66250	3.285	0.05000	25.2	-0.055	-1.02
1209	1.71250	3.237	0.05000	25.2	-0.048	-0.91
1239	1.76250	3.193	0.05000	25.2	-0.044	-0.83
1269	1.81250	3.153	0.05000	25.2	-0.040	-0.76
1300	1.86250	3.117	0.05000	25.2	-0.036	
1330	1.91250	3.083	0.05000	25.2	-0.034	
1360	1.96250	3.052	0.05000	25.2	-0.031	
1390	2.01250	3.023	0.05000	25.2	-0.029	
1420	2.06250	2.996	0.05000	25.2	-0.027	

Used Instruments

Name	SERCtitrate	Serial number	B433917554
------	-------------	---------------	------------

Responsible User

Time	Login name	Full name
2/22/2024 12:10:04 PM	SystemInternal	System internal account

- (1) Modified
- (9) Standard evaluation used
- (12) Unfinished sample
- (22) New (Manual)